

CHAPTER

5

Mathematical Induction
and Binomial Theorem

Section-A

JEE Advanced/ IIT-JEE

A Fill in the Blanks

- The larger of $99^{50} + 100^{50}$ and 101^{50} is
(1982 - 2 Marks)
- The sum of the coefficients of the polynomial $(1 + x - 3x^2)^{2163}$ is
(1982 - 2 Marks)
- If $(1 + ax)^n = 1 + 8x + 24x^2 + \dots$ then $a = \dots$ and $n = \dots$
(1983 - 2 Marks)
- Let n be positive integer. If the coefficients of 2nd, 3rd, and 4th terms in the expansion of $(1 + x)^n$ are in A.P., then the value of n is
(1994 - 2 Marks)
- The sum of the rational terms in the expansion of $(\sqrt{2} + 3^{1/5})^{10}$ is
(1997 - 2 Marks)

C MCQs with One Correct Answer

- Given positive integers $r > 1, n > 2$ and that the coefficient of $(3r)$ th and $(r + 2)$ th terms in the binomial expansion of $(1 + x)^{2n}$ are equal. Then
(1983 - 1 Mark)
(a) $n = 2r$ (c) $n = 2r + 1$
(b) $n = 3r$ (d) none of these
- The coefficient of x^4 in $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$ is (1983 - 1 Mark)
(a) $\frac{405}{256}$ (b) $\frac{504}{259}$
(c) $\frac{450}{263}$ (d) none of these
- The expression $\left(x + (x^3 - 1)^{\frac{1}{2}}\right)^5 + \left(x - (x^3 - 1)^{\frac{1}{2}}\right)^5$ is a polynomial of degree (1992 - 2 Marks)
(a) 5 (b) 6 (c) 7 (d) 8
- If in the expansion of $(1 + x)^m (1 - x)^n$, the coefficients of x and x^2 are 3 and -6 respectively, then m is (1999 - 2 Marks)
(a) 6 (b) 9 (c) 12 (d) 24
- For $2 \leq r \leq n$, $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2} =$ (2000S)
(a) $\binom{n+1}{r-1}$ (b) $2\binom{n+1}{r+1}$ (c) $2\binom{n+2}{r}$ (d) $\binom{n+2}{r}$

- In the binomial expansion of $(a - b)^n, n \geq 5$, the sum of the 5th and 6th terms is zero. Then a/b equals (2001S)
(a) $(n-5)/6$ (b) $(n-4)/5$
(c) $5/(n-4)$ (d) $6/(n-5)$

- The sum $\sum_{i=0}^m \binom{10}{i} \binom{20}{m-i}$, (where $\binom{p}{q} = 0$ if $p < q$) is maximum when m is (2002S)
(a) 5 (b) 10 (c) 15 (d) 20
- Coefficient of t^{24} in $(1+t^2)^{12} (1+t^{12}) (1+t^{24})$ is (2003S)
(a) ${}^{12}C_6 + 3$ (b) ${}^{12}C_6 + 1$ (c) ${}^{12}C_6$ (d) ${}^{12}C_6 + 2$
- If ${}^{n-1}C_r = (k^2 - 3) {}^nC_{r+1}$, then $k \in$ (2004S)
(a) $(-\infty, -2]$ (b) $[2, \infty)$ (c) $[-\sqrt{3}, \sqrt{3}]$ (d) $(\sqrt{3}, 2]$
- The value of

$$\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$$

is where

$$\binom{n}{r} = {}^nC_r \quad (2005S)$$

- (a) $\binom{30}{10}$ (b) $\binom{30}{15}$ (c) $\binom{60}{30}$ (d) $\binom{31}{10}$
- For $r = 0, 1, \dots, 10$, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}, (2010)$
 $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to
(a) $B_{10} - C_{10}$ (b) $A_{10}(B_{10}^2 - C_{10}A_{10})$
(c) 0 (d) $C_{10} - B_{10}$
- Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is (JEE Adv. 2014)
(a) 1051 (b) 1106 (c) 1113 (d) 1120

D MCQs with One or More than One Correct

- If C_r stands for nC_r , then the sum of the series
$$\frac{2\left(\frac{n}{2}\right)! \left(\frac{n}{2}\right)!}{n!} [C_0^2 - 2C_1^2 + 3C_2^2 - \dots + (-1)^n (n+1)C_n^2],$$

where n is an even positive integer, is equal to (1986 - 2 Marks)

(a) 0 (b) $(-1)^{n/2}(n+1)$

(c) $(-1)^{n/2}(n+2)$ (d) $(-1)^n n$

(e) none of these.

2. If $a_n = \sum_{r=0}^n \frac{1}{n C_r}$, then $\sum_{r=0}^n \frac{r}{n C_r}$ equals (1998 - 2 Marks)

(a) $(n-1)a_n$

(b) na_n

(c) $\frac{1}{2}na_n$

(d) None of the above

E Subjective Problems

1. Given that (1979)
 $C_1 + 2C_2x + 3C_3x^2 + \dots + 2nC_{2n}x^{2n-1} = 2n(1+x)^{2n-1}$

where $C_r = \frac{(2n)!}{r!(2n-r)!}$ $r=0, 1, 2, \dots, 2n$

Prove that

$C_1^2 - 2C_2^2 + 3C_3^2 - \dots - 2nC_{2n}^2 = (-1)^n C_n$

2. Prove that $7^{2n} + (2^{3n-3})(3^{n-1})$ is divisible by 25 for any natural number n . (1982 - 5 Marks)

3. If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ then show that the sum of the products of the C_i 's taken two at a time,

represented by $\sum_{0 \leq i < j \leq n} C_i C_j$ is equal to $2^{2n-1} - \frac{(2n)!}{2(n!)^2}$

(1983 - 3 Marks)

4. Use mathematical Induction to prove : If n is any odd positive integer, then $n(n^2 - 1)$ is divisible by 24.

(1983 - 2 Marks)

5. If p be a natural number then prove that $p^{n+1} + (p+1)^{2n-1}$ is divisible by $p^2 + p + 1$ for every positive integer n .

(1984 - 4 Marks)

6. Given $s_n = 1 + q + q^2 + \dots + q^n$;

$S_n = 1 + \frac{q+1}{2} + \left(\frac{q+1}{2}\right)^2 + \dots + \left(\frac{q+1}{2}\right)^n, q \neq 1$ Prove that

$n^{n+1}C_1 + n^{n+1}C_2s_1 + n^{n+1}C_3s_2 + \dots + n^{n+1}C_ns_n = 2^n S_n$

(1984 - 4 Marks)

7. Use method of mathematical induction $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24 for all $n > 0$ (1985 - 5 Marks)

8. Prove by mathematical induction that - (1987 - 3 Marks)

$\frac{(2n)!}{2^{2n}(n!)^2} \leq \frac{1}{(3n+1)^{1/2}}$ for all positive Integers n .

9. Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$, where $[]$ denotes the greatest integer function. Prove that $Rf = 4^{2n+4}$.

(1988 - 5 Marks)

10. Using mathematical induction, prove that (1989 - 3 Marks)

${}^m C_0 {}^n C_k + {}^m C_1 {}^n C_{k-1} + \dots + {}^m C_k {}^n C_0 = ({}^{m+n} C_k)$,

where m, n, k are positive integers, and ${}^p C_q = 0$ for $p < q$.

11. Prove that (1989 - 5 Marks)

$C_0 - 2^2 C_1 + 3^2 C_2 - \dots + (-1)^n (n+1)^2 C_n = 0,$

 $n > 2$, where $C_r = {}^n C_r$.

12. Prove that $\frac{n^7}{7} + \frac{n^5}{5} + \frac{2n^3}{3} - \frac{n}{105}$ is an integer for every positive integer n . (1990 - 2 Marks)

13. Using induction or otherwise, prove that for any non-negative integers m, n, r and k , (1991 - 4 Marks)

$$\sum_{m=0}^k (n-m) \frac{(r+m)!}{m!} = \frac{(r+k+1)!}{k!} \left[\frac{n}{r+1} - \frac{k}{r+2} \right]$$

14. If $\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r$ and $a_k = 1$ for all

$k \geq n$, then show that $b_n = {}^{2n+1} C_{n+1}$ (1992 - 6 Marks)

15. Let $p \geq 3$ be an integer and α, β be the roots of $x^2 - (p+1)x + 1 = 0$ using mathematical induction show that $\alpha^n + \beta^n$.

(i) is an integer and (ii) is not divisible by p (1992 - 6 Marks)

16. Using mathematical induction, prove that

$\tan^{-1}(1/3) + \tan^{-1}(1/7) + \dots + \tan^{-1}\{1/(n^2 + n + 1)\}$

$= \tan^{-1}\{n/(n+2)\}$ (1993 - 5 Marks)

17. Prove that $\sum_{r=1}^k (-3)^{r-1} {}^{3n} C_{2r-1} = 0$, where $k = (3n)/2$ and n is an even positive integer. (1993 - 5 Marks)

18. If x is not an integral multiple of 2π use mathematical induction to prove that : (1994 - 4 Marks)

$\cos x + \cos 2x + \dots + \cos nx = \cos \frac{n+1}{2} x \sin \frac{nx}{2} \operatorname{cosec} \frac{x}{2}$

19. Let n be a positive integer and (1994 - 5 Marks)
 $(1+x+x^2)^n = a_0 + a_1x + \dots + a_{2n}x^{2n}$
Show that $a_0^2 - a_1^2 + a_2^2 - \dots + a_{2n}^2 = a_n$

20. Using mathematical induction prove that for every integer $n \geq 1$, $(3^{2n}-1)$ is divisible by 2^{n+2} but not by 2^{n+3} . (1996 - 3 Marks)

21. Let $0 < A_i < \pi$ for $i = 1, 2, \dots, n$. Use mathematical induction to prove that

$\sin A_1 + \sin A_2 + \dots + \sin A_n \leq n \sin \left(\frac{A_1 + A_2 + \dots + A_n}{n} \right)$

where ≥ 1 is a natural number.

{You may use the fact that

$p \sin x + (1-p) \sin y \leq \sin [px + (1-p)y],$

where $0 \leq p \leq 1$ and $0 \leq x, y \leq \pi$.} (1997 - 5 Marks)

22. Let p be a prime and m a positive integer. By mathematical induction on m , or otherwise, prove that whenever r is an integer such that p does not divide r , p divides ${}^m p C_r$. (1998 - 8 Marks)

[Hint: You may use the fact that $(1+x)^{(m+1)p} = (1+x)^p (1+x)^{mp}$]

Mathematical Induction and Binomial Theorem

23. Let n be any positive integer. Prove that
(1999 - 10 Marks)

$$\sum_{k=0}^m \frac{\binom{2n-k}{k}}{\binom{2n-k}{n}} \cdot \frac{(2n-4k+1)}{(2n-2k+1)} 2^{n-2k} = \frac{\binom{n}{m}}{\binom{2n-2m}{n-m}} 2^{n-2m}$$

for each non-negative integer $m \leq n$. (Here $\binom{p}{q} = {}^pC_q$).

24. For any positive integer m, n (with $n \geq m$), let $\binom{n}{m} = {}^nC_m$.

$$\text{Prove that } \binom{n}{m} + \binom{n-1}{m} + \binom{n-2}{m} + \dots + \binom{m}{m} = \binom{n+1}{m+1}.$$

Hence or otherwise, prove that

$$\binom{n}{m} + 2\binom{n-1}{m} + 3\binom{n-2}{m} + \dots + (n-m+1)\binom{m}{m} = \binom{n+2}{m+2}.$$

(2000 - 6 Marks)

25. For every positive integer n , prove that

$$\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}. \text{ Hence or otherwise,}$$

prove that $[\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}]$, where $[x]$ denotes the greatest integer not exceeding x . (2000 - 6 Marks)

26. Let a, b, c be positive real numbers such that $b^2 - 4ac > 0$ and let $\alpha_1 = c$. Prove by induction that

$$\alpha_{n+1} = \frac{a\alpha_n^2}{(b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_n))} \text{ is well-defined and}$$

$$\alpha_{n+1} < \frac{\alpha_n}{2} \text{ for all } n = 1, 2, \dots \text{ (Here, 'well-defined' means}$$

that the denominator in the expression for α_{n+1} is not zero.)
(2001 - 5 Marks)

27. Use mathematical induction to show that
(25)ⁿ⁺¹ - 24n + 5735 is divisible by (24)² for all $n = 1, 2, \dots$
(2002 - 5 Marks)
28. Prove that
(2003 - 2 Marks)

$$2^k \binom{n}{0} \binom{n}{k} - 2^{k-1} \binom{n}{2} \binom{n}{k-1} \binom{n-1}{1} \binom{n-1}{k-1} \\ + 2^{k-2} \binom{n-2}{k-2} - \dots - (-1)^k \binom{n}{k} \binom{n-k}{0} = \binom{n}{k}.$$

29. A coin has probability p of showing head when tossed. It is tossed n times. Let p_n denote the probability that no two (or more) consecutive heads occur. Prove that $p_1 = 1$, $p_2 = 1 - p^2$ and $p_n = (1 - p) \cdot p_{n-1} + p(1 - p)p_{n-2}$ for all $n \geq 3$.

Prove by induction on n , that $p_n = A\alpha^n + B\beta^n$ for all $n \geq 1$, where α and β are the roots of quadratic equation

$$x^2 - (1-p)x - p(1-p) = 0 \text{ and } A = \frac{p^2 + \beta - 1}{\alpha\beta - \alpha^2}, B = \frac{p^2 + \alpha - 1}{\alpha\beta - \beta^2}.$$

(2000 - 5 Marks)

I

Integer Value Correct Type

1. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$ (JEE Adv. 2013)
2. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1) {}^{51}C_3$ for some positive integer n . Then the value of n is (JEE Adv. 2016)

Section-B

JEE Main / AIEEE

1. The coefficients of x^p and x^q in the expansion of $(1+x)^{p+q}$ are [2002]
(a) equal
(b) equal with opposite signs
(c) reciprocals of each other
(d) none of these
2. If the sum of the coefficients in the expansion of $(a+b)^n$ is 4096, then the greatest coefficient in the expansion is [2002]
(a) 1594 (b) 792 (c) 924 (d) 2924
3. The positive integer just greater than $(1+0.0001)^{10000}$ is [2002]
(a) 4 (b) 5 (c) 2 (d) 3
4. r and n are positive integers $r > 1, n > 2$ and coefficient of $(r+2)^{\text{th}}$ term and $3r^{\text{th}}$ term in the expansion of $(1+x)^{2n}$ are equal, then n equals [2002]
(a) $3r$ (b) $3r+1$ (c) $2r$ (d) $2r+1$
5. If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \dots}}}$ having n radical signs then by methods of mathematical induction which is true [2002]
(a) $a_n > 7 \forall n \geq 1$ (b) $a_n < 7 \forall n \geq 1$
(c) $a_n < 4 \forall n \geq 1$ (d) $a_n < 3 \forall n \geq 1$
6. If x is positive, the first negative term in the expansion of $(1+x)^{27/5}$ is [2003]
(a) 6th term (b) 7th term (c) 5th term (d) 8th term.
7. The number of integral terms in the expansion of $(\sqrt{3} + \sqrt[5]{5})^{256}$ is [2003]
(a) 35 (b) 32 (c) 33 (d) 34
8. Let $S(K) = 1 + 3 + 5 + \dots + (2K-1) = 3 + K^2$. Then which of the following is true [2004]
(a) Principle of mathematical induction can be used to prove the formula
(b) $S(K) \Rightarrow S(K+1)$
(c) $S(K) \not\Rightarrow S(K+1)$
(d) $S(1)$ is correct
9. The coefficient of the middle term in the binomial expansion in powers of x of $(1+\alpha x)^4$ and of $(1-\alpha x)^6$ is the same if α equals [2004]
(a) $\frac{3}{5}$ (b) $\frac{10}{3}$ (c) $-\frac{3}{10}$ (d) $-\frac{5}{3}$

10. The coefficient of x^n in expansion of $(1+x)(1-x)^n$ is
 (a) $(-1)^{n-1}n$ (b) $(-1)^n(1-n)$ [2004]
 (c) $(-1)^{n-1}(n-1)^2$ (d) $(n-1)$
11. The value of ${}^{50}C_4 + \sum_{r=1}^6 {}^{56-r}C_3$ is [2005]
 (a) ${}^{55}C_4$ (b) ${}^{55}C_3$ (c) ${}^{56}C_3$ (d) ${}^{56}C_4$
12. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then which one of the following holds for all $n \geq 1$, by the principle of mathematical induction [2005]
 (a) $A^n = nA - (n-1)I$ (b) $A^n = 2^{n-1}A - (n-1)I$
 (c) $A^n = nA + (n-1)I$ (d) $A^n = 2^{n-1}A + (n-1)I$
13. If the coefficient of x^7 in $\left[ax^2 + \left(\frac{1}{bx}\right)\right]^{11}$ equals the coefficient of x^{-7} in $\left[ax - \left(\frac{1}{bx^2}\right)\right]^{11}$, then a and b satisfy the relation [2005]
 (a) $a-b=1$ (b) $a+b=1$
 (c) $\frac{a}{b}=1$ (d) $ab=1$
14. If x is so small that x^3 and higher powers of x may be neglected, then $\frac{(1+x)^{\frac{3}{2}} - \left(1 + \frac{1}{2}x\right)^3}{(1-x)^{\frac{1}{2}}}$ may be approximated as [2005]
 (a) $1 - \frac{3}{8}x^2$ (b) $3x + \frac{3}{8}x^2$
 (c) $-\frac{3}{8}x^2$ (d) $\frac{x}{2} - \frac{3}{8}x^2$
15. If the expansion in powers of x of the function $\frac{1}{(1-ax)(1-bx)}$ is $a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, then a_n is [2006]
 (a) $\frac{b^n - a^n}{b-a}$ (b) $\frac{a^n - b^n}{b-a}$
 (c) $\frac{a^{n+1} - b^{n+1}}{b-a}$ (d) $\frac{b^{n+1} - a^{n+1}}{b-a}$
16. For natural numbers m, n if $(1-y)^m(1+y)^n = 1 + a_1y + a_2y^2 + \dots$ and $a_1 = a_2 = 10$, then (m, n) is [2006]
 (a) (20, 45) (b) (35, 20)
 (c) (45, 35) (d) (35, 45)
17. In the binomial expansion of $(a-b)^n$, $n \geq 5$, the sum of 5th and 6th terms is zero, then a/b equals [2007]
 (a) $\frac{n-5}{6}$ (b) $\frac{n-4}{5}$ (c) $\frac{5}{n-4}$ (d) $\frac{6}{n-5}$
18. The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - \dots + {}^{20}C_{10}$ is [2007]
 (a) 0 (b) ${}^{20}C_{10}$ (c) $-{}^{20}C_{10}$ (d) $\frac{1}{2} {}^{20}C_{10}$
19. Statement -1 : $\sum_{r=0}^n (r+1) {}^nC_r = (n+2)2^{n-1}$. [2008]
 Statement-2 : $\sum_{r=0}^n (r+1) {}^nC_r x^r = (1+x)^n + nx(1+x)^{n-1}$.
 (a) Statement -1 is false, Statement-2 is true
 (b) Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1
 (c) Statement -1 is true, Statement-2 is true; Statement -2 is not a correct explanation for Statement-1
 (d) Statement -1 is true, Statement-2 is false
20. The remainder left out when $8^{2n} - (62)^{2n+1}$ is divided by 9 is : [2009]
 (a) 2 (b) 7 (c) 8 (d) 0
21. Let $S_1 = \sum_{j=1}^{10} j(j-1) {}^{10}C_j$, $S_2 = \sum_{j=1}^{10} j {}^{10}C_j$ and $S_3 = \sum_{j=1}^{10} j^2 {}^{10}C_j$.
 Statement-1 : $S_3 = 55 \times 2^9$.
 Statement-2 : $S_1 = 90 \times 2^8$ and $S_2 = 10 \times 2^8$. [2010]
 (a) Statement -1 is true, Statement-2 is true ; Statement -2 is not a correct explanation for Statement -1.
 (b) Statement -1 is true, Statement -2 is false.
 (c) Statement -1 is false, Statement -2 is true .
 (d) Statement -1 is true, Statement 2 is true ; Statement -2 is a correct explanation for Statement -1.
22. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is [2011]
 (a) -132 (b) -144 (c) 132 (d) 144
23. If n is a positive integer, then $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$ is : [2012]
 (a) an irrational number
 (b) an odd positive integer
 (c) an even positive integer
 (d) a rational number other than positive integers
24. The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}}\right)^{10}$ is [JEE M 2013]
 (a) 4 (b) 120 (c) 210 (d) 310
25. If the coefficients of x^3 and x^4 in the expansion of $(1+ax+bx^2)(1-2x)^{18}$ in powers of x are both zero, then (a, b) is equal to: [JEE M 2014]
 (a) $\left(14, \frac{272}{3}\right)$ (b) $\left(16, \frac{272}{3}\right)$ (c) $\left(16, \frac{251}{3}\right)$ (d) $\left(14, \frac{251}{3}\right)$
26. The sum of coefficients of integral power of x in the binomial expansion $(1-2\sqrt{x})^{50}$ is : [JEE M 2015]
 (a) $\frac{1}{2}(3^{50}-1)$ (b) $\frac{1}{2}(2^{50}+1)$
 (c) $\frac{1}{2}(3^{50}+1)$ (d) $\frac{1}{2}(3^{50})$
27. If the number of terms in the expansion of $\left(1 - \frac{2}{x} + \frac{4}{x^2}\right)^n$, $x \neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion, is : [JEE M 2016]
 (a) 243 (b) 729 (c) 64 (d) 2187

5

Mathematical Induction and Binomial Theorem

Section-A : JEE Advanced/ IIT-JEE

- | | | | | | | |
|----------|-----------------|--------|---------------|---------|---------|---------|
| A | 1. $(101)^{50}$ | 2. -1 | 3. $a=2, n=4$ | 4. 7 | 5. 41 | |
| C | 1. (a) | 2. (a) | 3. (c) | 4. (c) | 5. (d) | 6. (b) |
| | 7. (c) | 8. (d) | 9. (d) | 10. (a) | 11. (d) | 12. (c) |
| D | 1. (c) | 2. (c) | | | | |
| I | 1. 6 | 2. 5 | | | | |

Section-B : JEE Main/ AIEEE

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (c) | 3. (d) | 4. (c) | 5. (b) | 6. (d) |
| 7. (c) | 8. (b) | 9. (c) | 10. (b) | 11. (d) | 12. (a) |
| 13. (d) | 14. (c) | 15. (d) | 16. (d) | 17. (b) | 18. (d) |
| 19. (b) | 20. (a) | 21. (b) | 22. (b) | 23. (a) | 24. (c) |
| 25. (b) | 26. (c) | 27. (b) | | | |

Section-A JEE Advanced/ IIT-JEE

A. Fill in the Blanks

- Consider $(101)^{50} - \{(99)^{50} + (100)^{50}\}$
 $= (100+1)^{50} - (100-1)^{50} - (100)^{50}$
 $= (100)^{50} [(1+0.01)^{50} - (1-0.01)^{50} - 1]$
 $= (100)^{50} [2({}^{50}C_1(0.01) + {}^{50}C_3(0.01)^3 + \dots) - 1]$
 $= (100)^{50} [2({}^{50}C_3(0.01)^3 + \dots)] > 0$
 $\therefore (101)^{50} > (99)^{50} + (100)^{50} \therefore (101)^{50}$ is greater.
- If we put $x=1$ in the expansion of $(1+x-3x^2)^{2163} = A_0 + A_1x + A_2x^2 + \dots$ we will get the sum of coefficients of given polynomial, which clearly comes to be -1.
- $(1+ax)^n = 1 + 8x + 24x^2 + \dots$

$$\Rightarrow (1+ax)^n = 1 + nax + \frac{n(n-1)}{2!} a^2 x^2 + \dots$$

$$= 1 + 8x + 24x^2 + \dots$$

Comparing like powers of x we get
 $nax = 8x \Rightarrow na = 8 \quad \dots(1)$

$$\frac{n(n-1)a^2}{2} = 24 \Rightarrow n(n-1)a^2 = 48 \quad \dots(2)$$

Solving (1) and (2), $n=4, a=2$

- We know that for a +ve integer n
 $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$
 ATQ coefficients of 2^{nd} , 3^{rd} , and 4^{th} terms are in A.P.
 i.e. ${}^nC_1, {}^nC_2, {}^nC_3$ are in A.P.
 $\Rightarrow 2 \cdot {}^nC_2 = {}^nC_1 + {}^nC_3$

$$\Rightarrow 2 \times \frac{n(n-1)}{2} = n + \frac{n(n-1)(n-2)}{3!}$$

$$\Rightarrow n-1 = 1 + \frac{n^2 - 3n + 2}{6} \Rightarrow n^2 - 9n + 14 = 0$$

$$\Rightarrow (n-7)(n-2) = 0 \Rightarrow n=7 \text{ or } 2$$

But for the existence of 4^{th} term, $n=7$.

- Let T_{r+1} be the general term in the expansion of $(\sqrt{2} + 3^{1/5})^{10}$

$$\therefore T_{r+1} = {}^{10}C_r (\sqrt{2})^{10-r} (3^{1/5})^r \quad (0 \leq r \leq 10)$$

$$= \frac{10!}{r!(10-r)!} 2^{5-r/2} 3^{r/5}$$

Let T_{r+1} will be rational if $2^{5-r/2}$ and $3^{r/5}$ are rational numbers.

$$\Rightarrow 5 - \frac{r}{2} \text{ and } \frac{r}{5} \text{ are integers.}$$

$$\Rightarrow r=0 \text{ and } r=10 \Rightarrow T_1 \text{ and } T_{11} \text{ are rational terms.}$$

$$\Rightarrow \text{Sum of } T_1 \text{ and } T_{11} = {}^{10}C_0 2^5 \cdot 3^0 + {}^{10}C_{10} 2^{5-5} \cdot 3^2$$

$$= 1.32.1 + 1.1.9 = 32 + 9 = 41$$

C. MCQs with ONE Correct Answer

- (a) Given that r and n are +ve integers such that $r > 1, n > 2$
 Also in the expansion of $(1+x)^{2n}$
 coeff. of $(3r)^{th}$ term = coeff. of $(r+2)^{th}$ term
 $\Rightarrow {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$
 $\Rightarrow 3r-1 = r+1 \text{ or } 3r-1+r+1 = 2n$
 $\Rightarrow r=1 \text{ or } 2r=n$
 But $r > 1 \therefore n=2r$

- (a) General term in the expansion $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$ is

$$T_{r+1} = {}^{10}C_r \left(\frac{x}{2}\right)^{10-r} \left(\frac{-3}{x^2}\right)^r = {}^{10}C_r x^{10-3r} \frac{(-1)^r 3^r}{2^{10-r}}$$

For coeff of x^4 , we should have

$$10 - 3r = 4 \Rightarrow r = 2$$

$$\therefore \text{Coeff of } x^4 = {}^{10}C_2 \frac{(-1)^2 3^2}{2^8} = \frac{405}{256}$$

3. (c) The given expression is

$$(x + \sqrt{x^3 - 1})^5 + (x - \sqrt{x^3 - 1})^5$$

We know by binomial theorem, that

$$(x + a)^n + (x - a)^n = 2 [{}^nC_0 x^n + {}^nC_2 x^{n-2} a^2 + {}^nC_4 x^{n-4} a^4 + \dots]$$

$\therefore$ The given expression is equal to

$$2 [{}^5C_0 x^5 + {}^5C_2 x^3 (x^3 - 1) + {}^5C_4 x (x^3 - 1)^2]$$

Max. power of x involved here is 7, also only +ve integral powers of x are involved, therefore given expression is a polynomial of degree 7.

4. (c) We have $(1+x)^m (1-x)^n$

$$\left[1 + mx + \frac{m(m-1)}{2!} x^2 + \dots \right] \left[1 - nx + \frac{n(n-1)}{2!} x^2 - \dots \right]$$

$$= 1 + (m-n)x + \left[\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn \right] x^2 + \dots$$

$$\text{Given, } m-n=3 \quad \dots(1)$$

$$\text{and } \frac{1}{2} m(m-1) + \frac{1}{2} n(n-1) - mn = -6$$

$$\Rightarrow m^2 + n^2 - 2mn - (m+n) = -12$$

$$\Rightarrow (m-n)^2 - (m+n) = -12$$

$$\Rightarrow m+n = 9+12 = 21 \quad \dots(2)$$

From (1) and (2), we get $m = 12$

5. (d) $\binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}$

$$= \left[\binom{n}{r} + \binom{n}{r-1} \right] + \left[\binom{n}{r-1} + \binom{n}{r-2} \right]$$

$$\text{NOTE THIS STEP: } \binom{n+1}{r} + \binom{n+1}{r-1} = \binom{n+2}{r}$$

$$[\therefore {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r]$$

6. (b) $(a-b)^n, n \geq 5$

In binomial expansion of above $T_5 + T_6 = 0$

$$\Rightarrow {}^nC_4 a^{n-4} b^4 + {}^nC_5 a^{n-5} b^5 = 0$$

$$\Rightarrow \frac{{}^nC_4}{{}^nC_5} \cdot \frac{a}{b} = 1 \Rightarrow \frac{4+1}{n-4} \cdot \frac{a}{b} = 1 \Rightarrow \frac{a}{b} = \frac{n-4}{5}$$

7. (c) $\sum_{i=0}^m {}^{10}C_i {}^{20}C_{m-i} = {}^{10}C_0 {}^{20}C_m + {}^{10}C_1 {}^{20}C_{m-1} + {}^{10}C_2 {}^{20}C_{m-2} + \dots + {}^{10}C_m {}^{20}C_0$
 $= \text{Coeff of } x^m \text{ in the expansion of product } (1+x)^{10} (1+x)^{20}$
 $= \text{Coeff of } x^m \text{ in the expansion of } (1+x)^{30}$
 $= {}^{30}C_m$

To get max. value of given sum, ${}^{30}C_m$ should be max. which is so when $m = 30/2 = 15$.

$$\left[\text{Using the fact that } \max ({}^nC_r) = \begin{cases} {}^nC_{n/2} & \text{if } n \text{ is even} \\ {}^nC_{\frac{n+1}{2}} & \text{if } n \text{ is odd} \end{cases} \right]$$

8. (d) $(1+t^2)^{12} (1+t^{12}) (1+t^{24})$
 $= (1+t^{12}+t^{24}+t^{36}) (1+t^2)^{12}$
 $\therefore \text{Coeff. of } t^{24} = 1 \times \text{Coeff. of } t^{24} \text{ in } (1+t^2)^{12} + 1 \times \text{Coeff. of } t^{12} \text{ in } (1+t^2)^{12} + 1 \times \text{constant term in } (1+t^2)^{12}$
 $= {}^{12}C_{12} + {}^{12}C_6 + {}^{12}C_0 = 1 + {}^{12}C_6 + 1 = {}^{12}C_6 + 2$

9. (d) ${}^{n-1}C_r = {}^nC_{r+1} (k^2 - 3) \Rightarrow k^2 - 3 = \frac{{}^{n-1}C_r}{{}^nC_{r+1}} = \frac{r+1}{n}$

Since $0 \leq r \leq n-1$

$$\Rightarrow 1 \leq r+1 \leq n \Rightarrow \frac{1}{n} \leq \frac{r+1}{n} \leq 1 \Rightarrow \frac{1}{n} \leq k^2 - 3 \leq 1$$

$$\Rightarrow 3 + \frac{1}{n} \leq k^2 \leq 4 \Rightarrow \sqrt{3 + \frac{1}{n}} \leq k \leq 2$$

$$\text{as } n \rightarrow \infty \Rightarrow \sqrt{3} < k \leq 2 \Rightarrow k \in (\sqrt{3}, 2]$$

10. (a) To find ${}^{30}C_0 {}^{30}C_{10} - {}^{30}C_1 {}^{30}C_{11} + {}^{30}C_2 {}^{30}C_{12} - \dots + {}^{30}C_{20} {}^{30}C_{30}$

We know that

$$(1+x)^{30} = {}^{30}C_0 + {}^{30}C_1 x + {}^{30}C_2 x^2 + \dots + {}^{30}C_{20} x^{20} + \dots + {}^{30}C_{30} x^{30} \quad \dots(1)$$

$$(x-1)^{30} = {}^{30}C_0 x^{30} - {}^{30}C_1 x^{29} + \dots + {}^{30}C_{10} x^{20} - \dots + {}^{30}C_{20} x^{10} - {}^{30}C_{30} x^0 \quad \dots(2)$$

Multiplying eqⁿ (1) and (2), we get

$$(x^2 - 1)^{30} = (\quad) \times (\quad)$$

$$\text{Equating the coefficients of } x^{20} \text{ on both sides, we get}$$

$${}^{30}C_{10} = {}^{30}C_0 {}^{30}C_{10} - {}^{30}C_1 {}^{30}C_{11} + {}^{30}C_2 {}^{30}C_{12} - \dots + {}^{30}C_{20} {}^{30}C_{30}$$

$\therefore$ Req. value is ${}^{30}C_{10}$

11. (d) Clearly $A_r = {}^{10}C_r, B_r = {}^{20}C_r, C_r = {}^{30}C_r$

$$\text{Now } \sum_{r=1}^{10} {}^{10}C_r ({}^{20}C_{10} {}^{20}C_r - {}^{30}C_{10} {}^{10}C_r)$$

$$= {}^{20}C_{10} \sum_{r=1}^{10} {}^{10}C_r {}^{20}C_r - {}^{30}C_{10} \sum_{r=1}^{10} {}^{10}C_r \times {}^{10}C_r$$

$$= {}^{20}C_{10} ({}^{10}C_1 {}^{20}C_1 + {}^{10}C_2 {}^{20}C_2 + \dots + {}^{10}C_{10} {}^{20}C_{10})$$

$$- {}^{30}C_{10} ({}^{10}C_1 \times {}^{10}C_1 + {}^{10}C_2 \times {}^{10}C_2 + \dots + {}^{10}C_{10} {}^{10}C_{10}) \dots(1)$$

Now expanding $(1+x)^{10}$ and $(1+x)^{20}$ by binomial theorem and comparing the coefficients of x^{20} in their product, on both sides, we get

$${}^{10}C_0 {}^{20}C_0 + {}^{10}C_1 {}^{20}C_1 + {}^{10}C_2 {}^{20}C_2 + \dots + {}^{10}C_{10} {}^{20}C_{10}$$

$$= \text{coeff of } x^{20} \text{ in } (1+x)^{30} = {}^{30}C_{20} = {}^{30}C_{10}$$

$$\therefore {}^{10}C_1 {}^{20}C_1 + {}^{10}C_2 {}^{20}C_2 + \dots + {}^{10}C_{10} {}^{20}C_{10} = {}^{30}C_{10} - 1$$

Again expanding $(1+x)^{10}$ and $(x+1)^{10}$ by binomial theorem and comparing the coefficients of x^{10} in their

product on both sides, we get

$$\therefore \left({}^{10}C_0\right)^2 \left({}^{10}C_1\right)^2 + \left({}^{10}C_2\right)^2 + \dots + \left({}^{10}C_{10}\right)^2 =$$

$$\text{coeff of } x^{10} \text{ in } (1+x)^{20} = {}^{20}C_{10}$$

$$\therefore \left({}^{10}C_1\right)^2 + \left({}^{10}C_2\right)^2 + \dots + \left({}^{10}C_{10}\right)^2 = {}^{20}C_{10} - 1$$

Substituting these values in equation (1), we get

$$= {}^{20}C_{10} \left({}^{30}C_{10} - 1\right) - {}^{30}C_{10} \left({}^{20}C_{10} - 1\right)$$

$$= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}$$

12. (c) Coeff. of x^{11} in exp. of $(1+x^2)^4 (1+x^3)^7 (1+x^4)^{12}$

$$= (\text{Coeff. of } x^a) \times (\text{Coeff. of } x^b) \times (\text{Coeff. of } x^c)$$

Such that $a + b + c = 11$

Here $a = 2m, b = 3n, c = 4p$

$$\therefore 2m + 3n + 4p = 11$$

Case I : $m = 0, n = 1, p = 2$

Case II : $m = 1, n = 3, p = 0$

Case III : $m = 2, n = 1, p = 1$

Case IV : $m = 4, n = 1, p = 0$

$\therefore$ Required coeff.

$$= {}^4C_0 \times {}^7C_1 \times {}^{12}C_2 + {}^4C_1 \times {}^7C_3 \times {}^{12}C_0 \\ + {}^4C_2 \times {}^7C_1 \times {}^{12}C_1 + {}^4C_4 \times {}^7C_1 \times {}^{12}C_0 \\ = 462 + 140 + 504 + 7 = 1113$$

D. MCQs with ONE or MORE THAN ONE Correct

1. (c) $\therefore n$ is even, let $n = 2m$ then

$$\text{LHL} = S = \frac{2.m!.m!}{(2m)!} [C_0^2 - 2C_1^2 + 3C_2^2 - \dots \\ + (-1)^{2m} (2m+1) C_{2m}^2] \dots (1) \\ = \frac{2.m!.m!}{(2m)!} C_{2m}^2 - 2C_{2m-1}^2 + 3C_{2m-2}^2 - \dots \\ + (-1)^{2m} (2m+1) C_0^2 \quad [\text{Using } C_r = C_{n-r}]$$

$$\Rightarrow S = \frac{2.m!.m!}{(2m)!} [(2m+1) C_0^2 - 2m C_1^2 \\ + (2m-1) C_2^2 - \dots - 2C_{2m-1}^2 + C_{2m}^2] \dots (2)$$

Adding (1) and (2):

$$2S = 2 \frac{m!.m!}{(2m)!} [2m+2] [C_0^2 - C_1^2 + C_2^2 + \dots + C_{2m}^2]$$

Now keeping in mind that if n is even, then

$$C_0^2 - C_1^2 + C_2^2 - \dots + C_n^2 = (-1)^{n/2} {}^nC_{n/2}$$

$\therefore$ we get

$$S = \frac{m!.m!}{(2m)!} (2m+2) [(-1)^m {}^{2m}C_m] = \left(2 \frac{n}{2} + 2\right) (-1)^{n/2} \\ = (-1)^{n/2} (n+2)$$

2. (c) Let $b = \sum_{r=0}^n \frac{r}{n C_r} = \sum_{r=0}^n \frac{n-r}{n C_r}$

$$= na_n - \sum_{r=0}^n \frac{n-r}{n C_{n-r}} \quad [\because {}^nC_r = {}^nC_{n-r}] \\ = na_n - b$$

$$\Rightarrow 2b = na_n \Rightarrow b = \frac{n}{2} a_n$$

E. Subjective Problems

1. Given that

$$C_1 + 2C_2x + 3C_3x^2 + \dots + 2nC_{2n}x^{2n-1} = 2n(1+x)^{2n-1} \dots (1)$$

where $C_r = \frac{2n!}{r!(2n-r)!}$

Integrating both sides with respect to x , under the limits 0 to x , we get

$$[C_1x + C_2x^2 + C_3x^3 + \dots + C_{2n}x^{2n}]_0^x = [(1+x)^{2n}]_0^x \\ \Rightarrow C_1x + C_2x^2 + C_3x^3 + \dots + C_{2n}x^{2n} = (1+x)^{2n} - 1 \\ \Rightarrow C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_{2n}x^{2n} = (1+x)^{2n} \dots (2)$$

Changing x by $-\frac{1}{x}$, we get

$$\Rightarrow C_0 - \frac{C_1}{x} + \frac{C_2}{x^2} - \frac{C_3}{x^3} + \dots + (-1)^{2n} \frac{C_{2n}}{x^{2n}} = \left(1 - \frac{1}{x}\right)^{2n} \\ \Rightarrow C_0x^{2n} - C_1x^{2n-1} + C_2x^{2n-2} - C_3x^{2n-3} \\ + \dots + C_{2n} = (x-1)^{2n} \dots (3)$$

Multiplying eqn. (1) and (3) and equating the coefficients of x^{2n-1} on both sides, we get

$$-C_1^2 + 2C_2^2 - 3C_3^2 + \dots + 2nC_{2n}^2 \\ = \text{coeff. of } x^{2n-1} \text{ in } 2n(x-1)(x^2-1)^{2n-1} \\ = 2n [\text{coeff. of } x^{2n-2} \text{ in } (x^2-1)^{2n-1} \\ - \text{coeff. of } x^{2n-1} \text{ in } (x^2-1)^{2n-1}] \\ = 2n [{}^{2n-1}C_{n-1}(-1)^{n-1} - 0] \\ = (-1)^{n-1} \cdot 2n {}^{2n-1}C_{n-1} \\ \Rightarrow C_1^2 - 2C_2^2 + 3C_3^2 + \dots + 2nC_{2n}^2 \\ = (-1)^n \cdot 2n {}^{2n-1}C_{n-1} = (-1)^n n \cdot \left(\frac{2n}{n} \cdot {}^{2n-1}C_{n-1}\right) \\ = (-1)^n n \cdot {}^{2n}C_n = (-1)^n n \cdot C_n \quad (\because {}^{2n}C_n = C_n)$$

Hence Proved.

2. $P(n) : 7^{2n} + 2^{3n-3} \cdot 3^{n-1}$ is divisible by 25 $\forall n \in \mathbb{N}$.

Let us prove it by Mathematical Induction :

$$P(1) : 7^2 + 2^0 \cdot 3^0 = 49 + 1 = 50 \text{ which is divisible by 25.}$$

$\therefore P(1)$ is true.

Let $P(k)$ be true that is $7^{2k} + 2^{3k-3} \cdot 3^{k-1}$ is divisible by 25.

$$\Rightarrow 7^{2k} + 2^{3k-3} \cdot 3^{k-1} = 25m \text{ where } m \in \mathbb{Z}.$$

$$\Rightarrow 2^{3k-3} \cdot 3^{k-1} = 25m - 7^{2k} \dots (1)$$

Consider $P(k+1)$:

$$7^{2(k+1)} + 2^{3(k+1)-3} \cdot 3^{k+1-1} = 7^{2k} \cdot 7^2 + 2^{3k} \cdot 3^k \\ = 49 \cdot 7^{2k} + 2^3 \cdot 3 \cdot 2^{3k-3} \cdot 3^{k-1} = 49 \cdot 7^{2k} + 24(25m - 7^{2k})$$

(Using IH eq. (1))

$$= 49 \cdot 7^{2k} + 24 \times 25m - 24 \times 7^{2k}$$

$$= 25 \cdot 7^{2k} + 24 \times 25m = 25(7^{2k} + 24m)$$

$$= 25 \times \text{some integral value which is divisible by 25.}$$

$\therefore P(k+1)$ is also true.

Hence by the principle of mathematical induction

$P(n)$ is true $\forall n \in \mathbb{Z}$.

$$3. \quad S = \sum_{i=0}^n \sum_{j=0}^n C_i C_j$$

$$0 \leq i < j \leq n$$

NOTE THIS STEP

$$\Rightarrow S = C_0(C_1 + C_2 + C_3 + \dots + C_n) + C_1(C_2 + C_3 + \dots + C_n) + C_2(C_3 + C_4 + C_5 + \dots + C_n) + \dots + C_{n-1}(C_n)$$

$$\Rightarrow S = C_0(2^n - C_0) + C_1(2^n - C_0 - C_1) + C_2(2^n - C_0 - C_1 - C_2) + \dots + C_{n-1}(2^n - C_0 - C_1 - \dots - C_{n-1})$$

$$\Rightarrow S = 2^n(C_0 + C_1 + C_2 + \dots + C_{n-1} + C_n) - (C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2) - S$$

$$\Rightarrow 2S = 2^n \cdot 2^n - \frac{2n!}{(n!)^2} = 2^{2n} - \frac{2n!}{(n!)^2}$$

$$\Rightarrow S = 2^{2n-1} - \frac{2n!}{2(n!)^2}$$

4. $P(n)$: $n(2^n - 1)$ is divisible by 24 for n odd +ve integer.

For $n = 2m - 1$, it can be restated as

$$P(m) : (2m - 1)(4m^2 - 4m) = 4m(m - 1)(2m - 1)$$

is divisible by 24 $\forall m \in \mathbb{N}$

$$\Rightarrow P(m) : m(m - 1)(2m - 1) \text{ is divisible by } 6 \quad \forall m \in \mathbb{N}.$$

Here $P(1) = 0$, divisible by 6.

$\therefore P(1)$ is true.

Let it be true for $m = k$, i.e.,

$$k(k-1)(2k-1) = 6p$$

$$\Rightarrow 2k^3 - 3k^2 + k = 6p \quad \dots(1)$$

Consider $P(k+1)$: $k(k+1)(2k+1) = 2k^3 + 3k^2 + k$

$$= 6p + 3k^2 + 3k^2 \quad (\text{Using (1)})$$

$$= 6(p + k^2) \Rightarrow \text{divisible by } 6$$

$\therefore P(k+1)$ is also true.

Hence $P(m)$ is true $\forall m \in \mathbb{N}$.

5. $P(n)$: $P^{n+1} + (p+1)^{2n-1}$ is divisible by $p^2 + p + 1$

For $n = 1$, $P(1)$: $p^2 + p + 1$ which is divisible by $p^2 + p + 1$.

$\therefore P(1)$ is true.

Let $P(k)$ be true, i.e.,

$$p^{k+1} + (p+1)^{2k-1} \text{ is divisible by } p^2 + p + 1$$

$$\Rightarrow p^{k+1} + (p+1)^{2k-1} = (p^2 + p + 1)m \quad \dots(1)$$

Consider $P(k+1)$: $p^{k+2} + (p+1)^{2k+1}$

$$= p \cdot p^{k+1} + (p+1)^{2k-1} \cdot (p+1)^2$$

$$= p[m(p^2 + p + 1) - (p+1)^{2k-1}] + (p+1)^{2k-1}(p+1)^2$$

$$= p(p^2 + p + 1)m - p(p+1)^{2k-1} + (p+1)^{2k-1}(p^2 + 2p + 1)$$

$$= p(p^2 + p + 1)m + (p+1)^{2k-1}(p^2 + p + 1)$$

$$= (p^2 + p + 1)[mp + (p+1)^{2k-1}]$$

$$= (p^2 + p + 1) \text{ some integral value}$$

$\therefore$ divisible by $p^2 + p + 1$ $\therefore P(k+1)$ is also true.

Hence by principle of mathematical induction $P(n)$ is true

$\forall n \in \mathbb{N}$.

$$6. \quad \text{We have } s_n = \frac{1 - q^{n+1}}{1 - q} \quad \dots(1)$$

$$\text{and } S_n = \frac{1 - \left(\frac{q+1}{2}\right)^{n+1}}{1 - \left(\frac{q+1}{2}\right)} = \frac{2^{n+1} - (q+1)^{n+1}}{2^n(1-q)} \quad \dots(2)$$

$$\text{Now, } {}^{n+1}C_1 + {}^{n+1}C_2 s_1 + {}^{n+1}C_3 s_2 + \dots + {}^{n+1}C_{n+1} s_n$$

$$= \frac{1}{1-q} [{}^{n+1}C_1(1-q) + {}^{n+1}C_2(1-q^2) + {}^{n+1}C_3(1-q^3) + \dots +$$

$$+ \dots + {}^{n+1}C_n(1-q^{n+1})] \quad \text{Using (1)}$$

$$= \frac{1}{1-q} \left[({}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_{n+1}) - ({}^{n+1}C_1 q + {}^{n+1}C_2 q^2 + \dots + {}^{n+1}C_{n+1} q^{n+1}) \right]$$

$$= \frac{1}{1-q} [2^{n+1} - 1 - \{(1+q)^{n+1} - 1\}]$$

$$= \frac{2^{n+1} - (1+q)^{n+1}}{(1-q)} = 2^n S_n \quad [\text{Using eq. (2)}]$$

7. Let $A_n = 2 \cdot 7^n + 3 \cdot 5^n - 5$

$$\text{Then } A_1 = 2 \cdot 7 + 3 \cdot 5 - 5 = 14 + 15 - 5 = 24.$$

Hence A_1 is divisible by 24.

Now assume that A_m is divisible by 24 so that we may write

$$A_m = 2 \cdot 7^m + 3 \cdot 5^m - 5 = 24k, \quad k \in \mathbb{N} \quad \dots(1)$$

$$\text{Then } A_{m+1} - A_m = 2(7^{m+1} - 7^m) + 3(5^{m+1} - 5^m) - 5 + 5$$

$$= 2 \cdot 7^m(7 - 1) + 3 \cdot 5^m(5 - 1) = 12 \cdot 7^m + 12 \cdot 5^m$$

Since 7^m and 5^m are odd integers $\forall m \in \mathbb{N}$, their sum must be an even integer, say $7^m + 5^m = 2p, p \in \mathbb{N}$.

$$\text{Hence } A_{m+1} - A_m = 12 \cdot 2p = 24p$$

$$\text{or } A_{m+1} = A_m + 24p = 24k + 24p \quad [\text{by (1)}]$$

Hence A_{m+1} is divisible by 24.

It follows by mathematical induction that A_n is divisible by 24 for all $n \in \mathbb{N}$.

$$8. \quad \text{Let } P(n) : \frac{(2n)!}{2^{2n}(n!)^2} \leq \frac{1}{(3n+1)^{1/2}}$$

$$\text{For } n=1, P(1) : \frac{2!}{2^2(1!)^2} \leq \frac{1}{(3+1)^{1/2}} \Rightarrow \frac{1}{4} \leq \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} \leq \frac{1}{2} \text{ which is true for } n=1$$

Assume that $P(k)$ is true, then

$$P(k) : \frac{(2k)!}{2^{2k}(k!)^2} \leq \frac{1}{(3k+1)^{1/2}} \quad \dots(1)$$

For $n = k+1$,

$$\frac{[2(k+1)]!}{2^{2(k+1)}[(k+1)!]^2} = \frac{(2k+2)!}{2^{2k+2}[(k+1)!]^2}$$

$$= \frac{(2k+2)(2k+1)(2k)!}{4 \cdot 2^{2k}(k+1)^2(k!)^2}$$

$$\leq \frac{(2k+2)(2k+1)}{4(k+1)^2} \cdot \frac{1}{(3k+1)^{1/2}}$$

[Using Induction hypothesis (1)]

$$= \frac{(2k+1)}{2 \cdot (k+1)(3k+1)^{1/2}}$$

$$\text{Thus, } \frac{[2(k+1)]!}{2^{2(k+1)}[(k+1)!]^2} \leq \frac{(2k+1)}{2(k+1)(3k+1)^{1/2}} \quad \dots(2)$$

Mathematical Induction and Binomial Theorem

In order to prove $P(k+1)$, it is sufficient to prove that

$$\frac{(2k+1)}{2(k+1)(3k+1)^{1/2}} \leq \frac{1}{(3k+4)^{1/2}} \quad \dots(3)$$

Squaring eq. (3), we get

$$\begin{aligned} \frac{(2k+1)^2}{4(k+1)^2(3k+1)} &\leq \frac{1}{3k+4} \\ \Rightarrow (2k+1)^2(3k+4) - 4(k+1)^2(3k+1) &\leq 0 \\ \Rightarrow (4k^2 + 4k + 1)(3k+4) - 4(k^2 + 2k + 1)(3k+1) &\leq 0 \\ \Rightarrow (12k^3 + 28k^2 + 19k + 4) - (12k^3 + 28k^2 + 20k + 4) &\leq 0 \\ \Rightarrow -k &\leq 0 \end{aligned}$$

which is true.

Hence from (2) and (3), we get

$$\frac{(2k+2)!}{2^{2k+2}[(k+1)!]^2} \leq \frac{1}{(3k+4)^{1/2}}$$

Hence the above inequation is true for $n = k+1$ and by the principle of induction it is true for all $n \in \mathbb{N}$.

9. We have $5\sqrt{5} - 11 = \frac{4}{5\sqrt{5} + 11} < 1$

Therefore $0 < 5\sqrt{5} - 11 < 1$

This gives us $0 < (5\sqrt{5} - 11)^{2n+1} < 1$ for every positive integer n .

$$\begin{aligned} \text{Also } (5\sqrt{5} + 11)^{2n+1} - (5\sqrt{5} - 11)^{2n+1} \\ = 2[{}^{2n+1}C_1(5\sqrt{5})^{2n} \cdot 11 + {}^{2n+1}C_3(5\sqrt{5})^{2n-2} \cdot 11^3 + \dots + {}^{2n+1}C_{2n+1}11^{2n+1}] \\ = 2[{}^{2n+1}C_1(125)^n \cdot 11 + {}^{2n+1}C_3(125)^{n-1} \cdot 11^3 + \dots + {}^{2n+1}C_{2n+1}11^{2n+1}] \\ = 2k \quad \dots(1) \end{aligned}$$

where k is some positive integer.

Let $F = (5\sqrt{5} - 11)^{2n+1}$

Then equation (1) becomes

$$\begin{aligned} R - F &= 2k \\ \Rightarrow [R] + R - [R] - F &= 2k \Rightarrow [R] + f - F = 2k \\ \Rightarrow f - F &= 2k - [R] \Rightarrow f - F \text{ is an integer.} \end{aligned}$$

But $0 \leq f < 1$ and $0 < F < 1$ Therefore $-1 < f - F < 1$

Since $f - F$ is an integer, we must have $f - F = 0$

$$\Rightarrow f = F.$$

$$\text{Now, } Rf = RF = (5\sqrt{5} + 11)^{2n+1}(5\sqrt{5} - 11)^{2n+1}$$

$$= [(5\sqrt{5})^2 - 12]^{2n+1} = 4^{2n+1}$$

10. Let the given statement be

$$P(m, n) : {}^mC_0 {}^nC_k + {}^mC_1 {}^nC_{k-1} + \dots + {}^mC_k {}^nC_0 = {}^{m+n}C_k$$

where $m, n, k \in \mathbb{N}$ and ${}^pC_q = 0$ for $p < q$.

As k is a positive integer and ${}^pC_q = 0$ for $p < q$.

$\therefore k$ must be a positive integer less than or equal to the smaller of m and n ,

We have $k = 1$, when $m = n = 1$

$$\therefore P(1, 1) \text{ is } {}^1C_0 {}^1C_1 + {}^1C_1 {}^1C_0 = {}^2C_1 \Rightarrow 1 + 1 = 2.$$

Thus $P(1, 1)$ is true.

Now let us assume that $P(m, n)$ holds good for any fixed value of m and n i.e.

$${}^mC_0 {}^nC_k + {}^mC_1 {}^nC_{k-1} + \dots + {}^mC_k {}^nC_0 = {}^{m+n}C_k \quad \dots(1)$$

Then $P(m+1, n+1)$ will be

$$\begin{aligned} {}^{m+1}C_0 {}^{n+1}C_k + {}^{m+1}C_1 {}^{n+1}C_{k-1} + \dots + {}^{m+1}C_k {}^{n+1}C_0 \\ = {}^{m+n+2}C_k \quad \dots(2) \end{aligned}$$

Consider LHS

$$\begin{aligned} &= {}^{m+1}C_0 {}^{n+1}C_k + {}^{m+1}C_1 {}^{n+1}C_{k-1} + \dots + {}^{m+1}C_k {}^{n+1}C_0 \\ &= 1 \cdot ({}^nC_{k-1} + {}^nC_k) + ({}^mC_0 + {}^mC_1)({}^nC_{k-2} + {}^nC_{k-1}) \\ &\quad + ({}^mC_1 + {}^mC_2)({}^nC_{k-3} + {}^nC_{k-2}) + \dots + ({}^mC_{k-1} + {}^mC_k) \cdot 1 \\ &= ({}^nC_{k-1} + {}^mC_1 {}^nC_{k-2} + {}^mC_2 {}^nC_{k-3} + \dots + {}^mC_{k-1} {}^nC_0) \\ &\quad + ({}^nC_k + {}^mC_1 {}^nC_{k-1} + {}^mC_2 {}^nC_{k-2} + \dots + {}^mC_{k-1} {}^nC_1 + {}^mC_k) \\ &\quad + ({}^mC_0 {}^nC_{k-2} + {}^mC_1 {}^nC_{k-3} + \dots + {}^mC_{k-2} {}^nC_0) \\ &\quad + ({}^mC_0 {}^nC_{k-1} + {}^mC_1 {}^nC_{k-2} + {}^mC_2 {}^nC_{k-3} \\ &\quad \quad \quad + \dots + {}^mC_{k-2} {}^nC_1 + {}^mC_{k-1}) \\ &= {}^{m+n}C_{k-1} + {}^{m+n}C_k + {}^{m+n}C_{k-2} + {}^{m+n}C_{k-1} \quad [\text{Using (1)}] \\ &= {}^{m+n+1}C_k + {}^{m+n+1}C_{k-1} = {}^{m+n+2}C_k \end{aligned}$$

Hence the theorem holds for the next integers $m+1$ and $n+1$. Then by mathematical induction the statement $P(m, n)$ holds for all positive integral values of m and n .

11. We know that

$$(1-x)^n = C_0 - C_1x + C_2x^2 - C_3x^3 + \dots + (-1)^n C_n x^n$$

Multiplying both sides by x , we get

$$x(1-x)^n = C_0x - C_1x^2 + C_2x^3 - C_3x^4 + \dots + (-1)^n C_n x^{n+1}$$

Differentiating both sides w.r. to x , we get

$$\begin{aligned} (1-x)^n - nx(1-x)^{n-1} \\ = C_0 - 2C_1x + 3C_2x^2 - 4C_3x^3 + \dots + (-1)^n (n+1) C_n x^n \end{aligned}$$

Again multiplying both sides by x , we get

$$\begin{aligned} x(1-x)^n - nx^2(1-x)^{n-1} \\ = C_0x - 2C_1x^2 + 3C_2x^3 - 4C_3x^4 + \dots + (-1)^n (n+1) C_n x^{n+1} \end{aligned}$$

Differentiating above with respect to x , we get

$$\begin{aligned} (1-x)^n - nx(1-x)^{n-1} - 2nx(1-x)^{n-1} + nx^2(n-1)(1-x)^{n-2} \\ = C_0 - 2^2C_1x + 3^2C_2x^2 - 4^2C_3x^3 + \dots + (-1)^n (n+1)^2 C_n x^n \end{aligned}$$

Substituting $x = 1$, in above, we get

$$0 = C_0 - 2^2C_1 + 3^2C_2 - 4^2C_3 + \dots + (-1)^n (n+1)^2 C_n$$

Hence Proved.

12. We have

$$P(n) : \frac{n^7}{7} + \frac{n^5}{5} + \frac{2n^3}{3} - \frac{n}{105} \text{ is an integer, } \forall n \in \mathbb{N}$$

$$P(1) : \frac{1}{7} + \frac{1}{5} + \frac{2}{3} - \frac{1}{105}$$

$$= \frac{15 + 21 + 70 - 1}{105} = \frac{105}{105} = 1 \text{ an integer}$$

$\therefore P(1)$ is true
Let $P(k)$ be true i.e.

$$\frac{k^7}{7} + \frac{k^5}{5} + \frac{2k^3}{3} - \frac{k}{105} \text{ is an integer}$$

$$\Rightarrow \frac{k^7}{7} + \frac{k^5}{5} + \frac{2k^3}{3} - \frac{k}{105} = m, (\text{say})$$

$$m \in \mathbb{N} \quad \dots(1)$$

Consider $P(k+1)$:

$$\begin{aligned} &= \frac{(k+1)^7}{7} + \frac{(k+1)^5}{5} + \frac{2(k+1)^3}{3} - \frac{(k+1)}{105} \\ &= \left(\frac{k^7 + 7k^6 + 21k^5 + 35k^4 + 35k^3 + 21k^2 + 7k + 1}{7} \right) \\ &\quad + \left(\frac{k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1}{5} \right) \\ &\quad + 2 \left(\frac{k^3 + 3k^2 + 3k + 1}{3} \right) - \left(\frac{k+1}{105} \right) \\ &= \left(\frac{k^7}{7} + \frac{k^5}{5} + \frac{2k^3}{3} - \frac{k}{105} \right) \\ &\quad + [k^6 + 3k^5 + 5k^4 + 5k^3 + 3k^2 + k + k^4 \\ &\quad + 2k^3 + 2k^2 + k + 2k^2 + 2k] + \left(\frac{1}{7} + \frac{1}{5} + \frac{2}{3} - \frac{1}{105} \right) \end{aligned}$$

$= m + \text{some integral value} + 1$

$= \text{some integral value}$

$\therefore P(k+1)$ is also true.

Hence $P(n)$ is true $\forall n \in \mathbb{N}$, (by the Principle of Mathematical Induction.)

13. Let $P(k) = \sum_{m=0}^k \frac{(n-m)(r+m)!}{m!} = \frac{(r+k+1)!}{k!} \left[\frac{n}{r+1} - \frac{k}{r+2} \right]$

For $k=1$, we will have two terms, on LHS, in sigma for $m=0$ and $m=1$, so that

$$LHS = (n-0) \frac{r!}{0!} + (n-1) \frac{(r+1)!}{1!}$$

$$\text{and } RHS = \frac{(r+2)!}{1!} \left[\frac{n}{r+1} - \frac{1}{r+2} \right]$$

Hence $LHS = RHS$ for $k=1$.

Now let the formula holds for $k=s$, that is let

$$\sum_{m=0}^s \frac{(n-m)(r+m)!}{m!} = \frac{(r+s+1)!}{s!} \left(\frac{n}{r+1} - \frac{s}{r+2} \right) \quad \dots(1)$$

Let us add next term corresponding to $m=s+1$ i.e.

adding $\frac{(n-s-1)(r+s+1)!}{(s+1)!}$ to both sides, we get

$$\begin{aligned} \sum_{m=0}^{s+1} \frac{(n-m)(r+m)!}{m!} &= \frac{(r+s+1)!}{s!} \left[\frac{n}{r+1} - \frac{s}{r+2} \right] \\ &\quad + \frac{(n-s-1)(r+s+1)!}{(s+1)!} \end{aligned}$$

$$\begin{aligned} &= \frac{(r+s+1)!}{(s+1)!} \left[\frac{(s+1)n}{r+1} - \frac{s(s+1)}{r+2} + n - s - 1 \right] \\ &= \frac{(r+s+1)!}{(s+1)!} \left[n \left\{ \frac{s+1}{r+1} + 1 \right\} - (s+1) \left\{ \frac{s}{r+2} + 1 \right\} \right] \\ &= \frac{(r+s+2)(r+s+1)!}{(s+1)!} \left[\frac{n}{r+1} - \frac{s+1}{r+2} \right] \end{aligned}$$

Hence the formula holds for $k=s+1$ and so by the induction principle, the formula holds for all natural numbers k .

14. Given that

$$\sum_{r=0}^{2n} a_r (x-2)^r = \sum_{r=0}^{2n} b_r (x-3)^r \quad \dots(1)$$

and $a_k = 1, \forall k \geq n$

To prove $b_n = {}^{2n+1}C_{n+1}$

In the given equation (1) let us put $x-3=y$ so that $x-2=y+1$ and we get

$$\begin{aligned} \sum_{r=0}^{2n} a_r (1+y)^r &= \sum_{r=0}^{2n} b_r (y)^r \\ \Rightarrow a_0 + a_1(1+y) + \dots + a_{n-1}(1+y)^{n-1} + (1+y)^n(1+y)^n \\ &\quad + (1+y)^{n+1} + \dots + (1+y)^{2n} \\ &= \sum_{r=0}^{2n} b_r y^r \quad [\text{Using } a_k = 1, \forall k \geq n] \end{aligned}$$

Equating the coefficients of y^n on both sides we get

NOTE THIS STEP:

$$\begin{aligned} \Rightarrow {}^nC_n + {}^{n+1}C_n + {}^{n+2}C_n + \dots + {}^{2n}C_n &= b_n \\ \Rightarrow ({}^{n+1}C_{n+1} + {}^{n+1}C_n) + {}^{n+2}C_n + \dots + {}^{2n}C_n &= b_n \\ &\quad [\text{Using } {}^nC_n = {}^{n+1}C_{n+1}] \\ \Rightarrow b_n &= {}^{n+2}C_{n+1} + {}^{n+2}C_n + \dots + {}^{2n}C_n \\ &\quad [\text{Using } {}^mC_r + {}^mC_{r-1} = {}^{m+1}C_r] \end{aligned}$$

Combining the terms in similar way, we get

$$\Rightarrow b_n = {}^{2n}C_{n+1} + {}^{2n}C_n \Rightarrow b_n = {}^{2n+1}C_{n+1}$$

Hence Proved

15. Since α, β are the roots of $x^2 - (p+1)x + 1 = 0$

$$\therefore \alpha + \beta = p+1; \alpha\beta = 1$$

Here $p \geq 3$ and $p \in \mathbb{Z}$

(i) To prove that $\alpha^n + \beta^n$ is an integer.

Let us consider the statement, " $\alpha^n + \beta^n$ is an integer."

Then for $n=1$, $\alpha + \beta = p+1$ which is an integer, p being an integer.

$\therefore$ Statement is true for $n=1$

Let the statement be true for $n \leq k$, i.e., $\alpha^k + \beta^k$ is an integer

Then,

$$\begin{aligned} \alpha^{k+1} + \beta^{k+1} &= \alpha^k \cdot \alpha + \beta^k \cdot \beta \\ &= \alpha(\alpha^k + \beta^k) + \beta(\alpha^k + \beta^k) - \alpha\beta^k - \alpha^k\beta \\ &= (\alpha + \beta)(\alpha^k + \beta^k) - \alpha\beta(\alpha^{k-1} + \beta^{k-1}) \\ &= (\alpha + \beta)(\alpha^k + \beta^k) - (\alpha^{k-1} + \beta^{k-1}) \quad \dots(1) \end{aligned}$$

[as $\alpha\beta = 1$]

= difference of two integers = some integral value

$\Rightarrow$ Statement is true for $n = k+1$.

$\therefore$ By the principle of mathematical induction the given statement is true for $\forall n \in \mathbb{N}$.

Mathematical Induction and Binomial Theorem

(ii) Let R_n be the remainder of $\alpha^n + \beta^n$ when divided by p

where $0 \leq R_n \leq p-1$

Since $\alpha + \beta = p+1 \therefore R_1 = 1$

$$\begin{aligned}\text{Also } \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta = (p+1)^2 - 2 \\ &= p^2 + 2p - 1 = p(p+1) + p - 1\end{aligned}$$

$\therefore R_2 = p-1$

Also from equation (1) of previous part (i), we have

$$\begin{aligned}\alpha^{n+1} + \beta^{n+1} &= (p+1)(\alpha^n + \beta^n) - (\alpha^{n-1} + \beta^{n-1}) \\ &= p(\alpha^n + \beta^n) + (\alpha^n + \beta^n) - (\alpha^{n-1} + \beta^{n-1})\end{aligned}$$

$\Rightarrow R_{n+1}$ is the remainder of $R_n - R_{n-1}$ when divided by p

$\therefore$ We observe that $R_2 - R_1 = p-1-1$

$\therefore R_3 = p-2$

Similarly, R_4 is the remainder when $R_3 - R_2$ is divided by p where

$$R_3 - R_2 = p-2 - p+1 = -1 = -p + (p-1) \therefore R_4 = p-1$$

$$R_4 - R_3 = p-1 - p+1 = 1 \therefore R_5 = 1$$

$$R_5 - R_4 = 1 - p+1 = -p+2 \therefore R_6 = p-2$$

It is evident for above that the remainder is either 1 or $p-1$ or $p-2$.

Since $p \geq 3$, so none is divisible by p .

16. To prove

$$\begin{aligned}P(n): \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \dots \tan^{-1}\left(\frac{1}{n^2 + n + 1}\right) \\ = \tan^{-1}\left(\frac{n}{n+2}\right)\end{aligned}$$

$$\text{For } n=1, \text{ LHS} = \tan^{-1}\frac{1}{3};$$

$$\text{RHS} = \tan^{-1}\frac{1}{3} \Rightarrow \text{LHS} = \text{RHS.}$$

$\therefore P(1)$ is true.

Let $P(k)$ be true, i.e.

$$\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \dots \tan^{-1}\left(\frac{1}{k^2 + k + 1}\right) = \tan^{-1}\left(\frac{k}{k+2}\right)$$

Consider $P(k+1)$

$$\begin{aligned}\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} + \dots \tan^{-1}\left(\frac{1}{k^2 + k + 1}\right) \\ + \tan^{-1}\left(\frac{1}{(k+1)^2 + (k+1) + 1}\right) \\ = \tan^{-1}\left[\frac{k+1}{(k+1)+2}\right]\end{aligned}$$

$$\begin{aligned}\text{LHS} &= \tan^{-1}\left[\frac{k}{k+2}\right] + \tan^{-1}\left(\frac{1}{k^2 + 3k + 3}\right) \\ &\quad \text{[Using equation (1)]}\end{aligned}$$

$$= \tan^{-1}\left[\frac{\frac{k}{k+2} + \frac{1}{k^2 + 3k + 3}}{1 - \left(\frac{k}{k+2}\right)\left(\frac{1}{k^2 + 3k + 3}\right)}\right]$$

$$= \tan^{-1}\left[\frac{(k+1)(k^2 + 2k + 2)}{(k+3)(k^2 + 2k + 2)}\right] = \tan^{-1}\left(\frac{k+1}{k+3}\right) = \text{RHS}$$

$\therefore P(k+1)$ is also true.

Hence by the principle of mathematical induction $P(n)$ is true for every natural number.

17. To evaluate $\sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1}$ where $k = \frac{3n}{2}$ and n is +ve even interger.

$$\text{Let } n=2m, \text{ where } m \in \mathbb{Z}^+ \therefore k = \frac{3(2m)}{2} = 3m$$

$$\begin{aligned}\therefore \sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1} &= \sum_{r=1}^{3m} (-3)^{r-1} {}^{6m}C_{2r-1} \\ &= {}^{6m}C_1 - 3.{}^{6m}C_3 + 3^2.{}^{6m}C_5 - \dots \dots \dots \dots (1)\end{aligned}$$

Now we know that

$$\begin{aligned}(1+a)^{6m} - (1-a)^{6m} \\ = 2[{}^{6m}C_1 a + {}^{6m}C_3 a^3 + {}^{6m}C_5 a^5 + \dots] \dots (2)\end{aligned}$$

Keeping in mind the form of RHS in equation (1) and in equation (2)

We put $a = i\sqrt{3}$ in equation (2) to get

$$\begin{aligned}(1+i\sqrt{3})^{6m} - (1-i\sqrt{3})^{6m} \\ = 2[{}^{6m}C_1 i\sqrt{3} - {}^{6m}C_3 i3\sqrt{3} + {}^{6m}C_5 i3^2\sqrt{3} \dots] \\ \Rightarrow (1+i\sqrt{3})^{6m} - (1-i\sqrt{3})^{6m} \\ = 2\sqrt{3}i[{}^{6m}C_1 - 3.{}^{6m}C_3 + 3^2.{}^{6m}C_5 \dots] \dots (3)\end{aligned}$$

$$\text{But } 1+i\sqrt{3} = 2(\cos \pi/3 + i \sin \pi/3)$$

$$\therefore (1+i\sqrt{3})^{6m} = 2^{6m}(\cos \pi/3 + i \sin \pi/3)^{6m}$$

NOTE THIS STEP

$$= 2^{6m}\left(\cos \frac{6m\pi}{3} + i \sin \frac{6m\pi}{3}\right) \text{ [Using D'Moivre's thm.]}$$

Similarly,

$$(1-i\sqrt{3})^{6m} = 2^{6m}\left(\cos \frac{6m\pi}{3} - i \sin \frac{6m\pi}{3}\right)$$

$$\therefore (1+i\sqrt{3})^{6m} - (1-i\sqrt{3})^{6m} = 2^{6m}.2 \sin 2m\pi = 0$$

Substituting the above in equation (3) we get

$${}^{6m}C_1 - 3.{}^{6m}C_3 + 3^2.{}^{6m}C_5 - \dots = 0$$

$$\Rightarrow \sum_{r=1}^k (-3)^{r-1} {}^{3n}C_{2r-1} = 0.$$

Hence Proved

18. Let $P(n) : \cos x + \cos 2x + \dots + \cos nx$

$$= \cos \frac{n+1}{2} x \sin \frac{nx}{2} \operatorname{cosec} \frac{x}{2} \dots (1)$$

where x is not an integral multiple of 2π .

For $n=1$ $P(1) : \text{L.H.S.} = \cos x$

$$\text{R.H.S.} = \cos \frac{1+1}{2} x \sin \frac{x}{2} \operatorname{cosec} \frac{x}{2} = \cos x$$

L.H.S. = R.H.S.

$\Rightarrow P(1)$ is true.

Let $P(k)$ be true i.e.

$$\cos x + \cos 2x + \dots + \cos kx$$

$$= \cos \frac{k+1}{2} x \sin \frac{kx}{2} \operatorname{cosec} \frac{x}{2} \quad \dots(2)$$

Consider $P(k+1)$:

$$\cos x + \cos 2x + \dots + \cos kx + \cos (k+1)x$$

$$= \cos \left(\frac{k+2}{2} \right) x \sin \frac{(k+1)x}{2} \operatorname{cosec} \frac{x}{2}$$

$$\text{L.H.S. } [\cos x + \cos 2x + \dots + \cos kx + \cos (k+1)x]$$

$$= \cos \left(\frac{k+1}{2} \right) x \sin \operatorname{cosec} \frac{kx}{2} \frac{x}{2} + \cos (k+1)x$$

[Using (2)]

$$= \left[\cos \left(\frac{k+1}{2} \right) x \sin \frac{kx}{2} + \cos (k+1)x \sin \frac{x}{2} \right] \operatorname{cosec} \frac{x}{2}$$

$$= \frac{1}{2} \left[2 \cos \frac{(k+1)x}{2} \sin \frac{kx}{2} + 2 \cos (k+1)x \sin \frac{x}{2} \right] \operatorname{cosec} \frac{x}{2}$$

$$= \frac{1}{2} \left[\sin \left(\frac{2k+1}{2} \right) x - \sin \frac{x}{2} \right. \\ \left. + \sin \left(xk + \frac{3x}{2} \right) - \sin \left(xk + \frac{x}{2} \right) \right] \operatorname{cosec} \frac{x}{2}$$

$$= \frac{1}{2} \left[\sin \left(xk + \frac{3x}{2} \right) - \sin \frac{x}{2} \right] \operatorname{cosec} \frac{x}{2}$$

$$= \frac{1}{2} \left[2 \cos \frac{(k+2)x}{2} \sin \frac{(k+1)x}{2} \right] \operatorname{cosec} \frac{x}{2}$$

$$= \cos \frac{(k+2)x}{2} \sin \frac{(k+1)x}{2} \operatorname{cosec} \frac{x}{2} = R.H.S.$$

$\therefore P(k+1)$ is also true.

Hence by the principle of mathematical induction

$P(n)$ is true $\forall n \in \mathbb{N}$.

19.

Given that,

$$(1+x+x^2)^n = a_0 + a_1x + \dots + a_{2n}x^{2n} \quad \dots(1)$$

where n is a +ve integer.

Replacing x by $-\frac{1}{x}$ in eqⁿ (1), we get

$$\left(1 - \frac{1}{x} + \frac{1}{x^2} \right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} - \frac{a_3}{x^3} + \dots + \frac{a_{2n}}{x^{2n}} \quad \dots(2)$$

Multiplying eq. (1) and (2):

$$\frac{(1+x+x^2)^n (x^2-x+1)^n}{x^{2n}}$$

$$= (a_0 + a_1x + \dots + a_{2n}x^{2n}) \left(a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} - \dots + \frac{a_n}{x^{2n}} \right)$$

Equating the constant terms on both sides we get

$$a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 = \text{constant term in the}$$

$$\text{expansion of } \frac{[(1+x+x^2)(1-x+x^2)]^n}{x^{2n}}$$

$$= \text{Coeff. of } x^{2n} \text{ in the expansion of } (1+x^2+x^4)^n$$

But replacing x by x^2 in eq's (1), we have

$$(1+x^2+x^4)^n = a_0 + a_1x^2 + \dots + a_{2n}(x^2)^{2n}$$

$$\therefore \text{Coeff of } x^{2n} = a_n$$

$$\text{Hence we obtain, } a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 = a_n$$

20. For $n=1$, $3^{2^1} - 1 = 3^2 - 1 = 9 - 1 = 8$ which is divisible by $2^{n+2} = 2^3 = 8$ but is not divisible by $2^{n+3} = 2^4 = 16$

Therefore, the result is true for $n=1$.

Assume that the result is true for $n=k$. That is, assume that

$$3^{2^k} - 1 \text{ is divisible by } 2^{k+2} \text{ but is not divisible by } 2^{k+3},$$

Since $3^{2^k} - 1$ is divisible by 2^{k+2} but not by 2^{k+3} , we can

$$\text{write } 3^{2^k} - 1 = (m) 2^{k+2}$$

where m must be an odd positive integer, for otherwise $3^{2^k} - 1$ will become divisible by 2^{k+3} .

$$\text{For } n=k+1, \text{ we have } 3^{2^{k+1}} - 1 = 3^{2^k \cdot 2} - 1 = (3^{2^k})^2 - 1$$

$$= (m \cdot 2^{k+2} + 1)^2 - 1 \quad [\text{Using (1)}]$$

$$= m^2 \cdot (2^{k+2})^2 + 2m \cdot 2^{k+2} + 1 - 1$$

$$= m^2 \cdot 2^{2k+4} + m \cdot 2^{k+3} = 2^{k+3} (m^2 \cdot 2^{k+1} + m)$$

$$\Rightarrow 3^{2^{k+1}} - 1 \text{ is divisible by } 2^{k+3}.$$

But $3^{2^{k+1}} - 1$ is not divisible by 2^{k+4} for otherwise we must have 2 divides $m^2 \cdot 2^{k+1} + m$. But this is not possible as m is odd. Thus, the result is true for $n=k+1$.

21. For $n=1$, the inequality becomes

$$\sin A_1 \leq \sin A_1, \text{ which is clearly true.}$$

Assume that the inequality holds for $n=k$ where k is some positive integer. That is, assume that

$$\sin A_1 + \sin A_2 + \dots + \sin A_k \leq k \sin \left(\frac{A_1 + A_2 + \dots + A_k}{k} \right) \quad \dots(1)$$

for same positive integer k .

We shall now show that the result holds for $n=k+1$ that is, we show that

$$\sin A_1 + \sin A_2 + \dots + \sin A_k + \sin A_{k+1} \\ \leq (k+1) \sin \left(\frac{A_1 + A_2 + \dots + A_{k+1}}{k+1} \right) \quad \dots(2)$$

L.H.S. of (2)

$$= \sin A_1 + \sin A_2 + \dots + \sin A_k + \sin A_{k+1}$$

$$\leq k \sin \left(\frac{A_1 + A_2 + \dots + A_k}{k} \right) + \sin A_{k+1}$$

[Induction assumption]

$$= (k+1) \left[\frac{k}{k+1} \sin \alpha + \frac{1}{k+1} \sin A_{k+1} \right];$$

$$\text{where } \alpha = \frac{A_1 + A_2 + \dots + A_k}{k}$$

$$\therefore \text{L.H.S. of (2)} \leq (k+1) \left[\left(1 - \frac{k}{k+1} \right) \sin \alpha + \frac{1}{k+1} \sin A_{k+1} \right]$$

$$\leq (k+1) \sin \left\{ \left(1 - \frac{k}{k+1} \right) \alpha + \frac{1}{k+1} A_{k+1} \right\}$$

$$[\text{Using the fact } p \sin x + (1-p) \sin y \leq \sin [px + (1-p)y] \text{ for } 0 \leq p \leq 1, 0 \leq x, y \leq \pi]$$

$$= (k+1) \sin \left\{ \frac{k}{k+1} \left(\frac{A_1 + A_2 + \dots + A_k}{k} \right) + \frac{1}{k+1} A_{k+1} \right\}$$

$$= (k+1) \sin \left(\frac{A_1 + A_2 + \dots + A_{k+1}}{k+1} \right)$$

Thus, the inequality holds for $n = k + 1$. Hence, by the principle of mathematical induction the inequality holds for all $n \in N$.

22. We know that ${}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1}$

$$\therefore {}^{mp}C_r = \frac{mp}{r} {}^{mp-1}C_{r-1} = \left[\frac{m \cdot {}^{mp-1}C_{r-1}}{r} \right] p$$

Now, L.H.S is an integer

$\Rightarrow$ RHS must be an integer

But p and r are coprime (given)

$$\therefore r \text{ must divide } m \cdot {}^{mp-1}C_{r-1}$$

$$\text{or } \frac{m \cdot {}^{mp-1}C_{r-1}}{r} \text{ is an integer.}$$

$$\Rightarrow \frac{{}^{mp}C_r}{p} \text{ is an integer or } {}^{mp}C_r \text{ is divisible by } p.$$

$$23. \text{ Let } P(m) = \sum_{k=0}^m \frac{\binom{2n-k}{k} (2n-4k+1)}{\binom{2n-k}{n}} 2^{n-2k}$$

$$= \frac{\binom{n}{m}}{\binom{2n-2m}{n-m}} 2^{n-2m} \quad \dots(1)$$

$$\text{For } m=0, \text{ LHS} = \frac{\binom{2n}{0}}{\binom{2n}{n}} \cdot \frac{2n+1}{2n+1} \cdot 2^n = \frac{1}{\binom{2n}{n}} 2^n,$$

$$\text{R.H.S.} = \frac{\binom{n}{0}}{\binom{2n}{n}} \cdot 2^n = \frac{1}{\binom{2n}{n}} 2^n = \text{L.H.S}$$

$$[\because m=0 \Rightarrow k=0]$$

$\therefore P(0)$ holds true. Now assuming $P(m)$

L.H.S. of $P(m+1) = \text{L.H.S. of}$

$$P(m) + \frac{\binom{2n-m-1}{m+1}}{\binom{2n-m-1}{n}} \cdot \frac{(2n-4m-3)}{(2n-2m-1)} 2^{n-2m-2}$$

$$= \frac{n!(n-m)!}{m!(2n-2m)!} 2^{n-2m}$$

$$+ \frac{n!(n-m-1)!(2n-4m-3)}{(m+1)!(2n-2m-2)!(2n-2m-1)} 2^{n-2m-2}$$

$$= \frac{n!(n-m-1)! 2^{n-2m-2}}{(m+1)!(2n-2m-1)!}$$

$$\times \left\{ \frac{(n-m) \cdot 4(m+1)}{(2n-2m)} + (2n-4m-3) \right\}$$

$$= \frac{n!(n-m-1)! 2^{n-2m-2} (2n-2m-1)}{(m+1)!(2n-2m-1)!}$$

$$= \frac{n!(n-m-1)! 2^{n-2m-2}}{(m+1)!(2n-2m-2)!} = \frac{\binom{n}{m+1}}{\binom{2n-2m-2}{n-m-1}} 2^{n-2m-2}$$

= R.H.S. of $P(m+1)$.

Hence by mathematical induction, result follows for all

$$0 \leq m \leq n.$$

24. Given that for positive integers m and n such that $n \geq m$, then to prove that

$${}^nC_m + {}^{n-1}C_m + {}^{n-2}C_m + \dots + {}^mC_m = {}^{n+1}C_{m+1}$$

L.H.S. ${}^nC_m + {}^{n-1}C_m + {}^{n-2}C_m + \dots + {}^mC_m = {}^{n+1}C_{m+1}$

[writing L.H.S. in reverse order]

$$= ({}^{m+1}C_{m+1} + {}^{m+1}C_m) + {}^{m+2}C_m + \dots + {}^{n-1}C_m + {}^nC_m$$

[$\because {}^mC_m = {}^{m+1}C_{m+1}$]

$$= ({}^{m+2}C_{m+1} + {}^{m+2}C_m) + {}^{m+3}C_m + \dots + {}^nC_m$$

[$\because {}^{n-1}C_m = {}^nC_m$]

$$= {}^{m+3}C_{m+1} + {}^{m+3}C_m + \dots + {}^nC_m$$

Combining in the same way we get

$$= {}^nC_{m+1} + {}^nC_m = {}^{n+1}C_{m+1} = \text{R.H.S.}$$

Again we have to prove

$${}^nC_m + 2{}^{n-1}C_m + 3{}^{n-2}C_m + \dots + (n-m+1){}^mC_m$$

$$= {}^{n+2}C_{m+2}$$

$$= [{}^nC_m + {}^{n-1}C_m + {}^{n-2}C_m + \dots + {}^mC_m] + [{}^{n-1}C_m + {}^{n-2}C_m + \dots + {}^mC_m] + \dots + [{}^mC_m]$$

[$n-m+1$ bracketed terms]

$$= {}^{n+1}C_{m+1} + {}^nC_{m+1} + {}^{n-1}C_{m+1} + \dots + {}^mC_{m+1}$$

[using previous result.]

$$= {}^{n+2}C_{m+2}$$

$$[\text{Replacing } n \text{ by } n+1 \text{ and } m \text{ by } m+1 \text{ in the previous result.}]$$

$$= \text{R.H.S.}$$

25. For $n > 0$, $\sqrt{4n+1} > 0$, $\sqrt{n} + \sqrt{n+1} > 0$ and $\sqrt{4n+2} > 0$

Now, $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$ to be proved.

I. To prove $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1}$

Squaring both sides in $\sqrt{4n+1} < \sqrt{n} + \sqrt{n+1}$

$$\Rightarrow 4n+1 < n+n+1+2\sqrt{n(n+1)}$$

$$\Rightarrow 2n < 2\sqrt{n(n+1)} \Rightarrow n < \sqrt{n(n+1)} \text{ which is true.}$$

II. To prove $\sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$

Squaring both sides,

$$n+n+1+2\sqrt{n(n+1)} < 4n+2$$

$$\Rightarrow 2\sqrt{n(n+1)} < 2n+1 \text{ Squaring again}$$

$$4[n(n+1)] < 4n^2+1+4n \text{ or } 0 < 1 \text{ which is true}$$

$$\text{Hence } \sqrt{4n+1} < \sqrt{n} + \sqrt{n+1} < \sqrt{4n+2}$$

Further to prove $[\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}]$, we have to prove that there is no positive integer which lies between $\sqrt{4n+1}$ and $\sqrt{4n+2}$ or $[\sqrt{4n+1}] = [\sqrt{4n+2}]$. Using Mathematical induction.

We have to check $[\sqrt{4n+1}] = [\sqrt{4n+2}]$ for $n = 1$

$[\sqrt{5}] = [\sqrt{6}] \Rightarrow 2 = 2$, which is true

Assume for $n = k$ (arbitrary)

i.e., $[\sqrt{4k+1}] = [\sqrt{4k+2}]$ To prove for $n = k+1$

To check $[\sqrt{4k+5}] = [\sqrt{4k+6}]$ since $k \geq 0$

Here $4k+5$ is an odd number and $4k+6$ is even number.

Their greatest integer will be different iff $4k+6$ is a perfect square that is $4k+6 = r^2$

$\Rightarrow k = \frac{r^2}{4} - \frac{6}{4}, \frac{6}{4}$ is not integer. But k has to be integer.

So $4k+6$ cannot be perfect square.

$\Rightarrow [\sqrt{4k+5}] = [\sqrt{4k+6}]$

By Sandwich theorem

$\Rightarrow [\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}]$

26. We have a, b, c the +ve real number s.t. $b^2 - 4ac > 0; \alpha_1 = c$.

$$P(n) : \alpha_{n+1} = \frac{a\alpha_n^2}{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_n)}$$

is well defined and $\alpha_{n+1} < \frac{\alpha_n}{2}, \forall n = 1, 2, \dots$

$$\text{For } n = 1, \alpha_2 = \frac{a\alpha_1^2}{b^2 - 2a\alpha_1} = \frac{ac^2}{b^2 - 2ac}$$

Now, $b^2 - 4ac > 0 \Rightarrow b^2 - 2ac > 2ac > 0$

$\therefore \alpha_2$ is well defined (as denomination is not zero)

$$\text{Also } \left[\begin{array}{l} \because b^2 - 2ac > 2ac \\ \Rightarrow \frac{1}{b^2 - 2ac} < \frac{1}{2ac} \end{array} \right] \Rightarrow \frac{\alpha_2}{c} < \frac{1}{2} \Rightarrow \frac{\alpha_2}{\alpha_1} < \frac{1}{2}$$

$\therefore P(n)$ is true for $n = 1$.

Let the statement be true for $1 \leq n \leq k$ i.e.,

$$\alpha_{k+1} = \frac{a\alpha_k^2}{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_k)} \text{ is well defined}$$

and $\alpha_{k+1} < \frac{\alpha_k}{2}$

Now, we will prove that $P(k+1)$ is also true

$$\text{i.e., } \alpha_{k+2} = \frac{a\alpha_{k+1}^2}{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_k + \alpha_{k+1})} \text{ is}$$

well defined and $\alpha_{k+2} < \frac{\alpha_{k+1}}{2}$.

We have

$$\alpha_1 = c, \alpha_2 < \frac{c}{2}, \alpha_3 < \frac{\alpha_2}{2} < \frac{c}{2^2}, \alpha_4 < \frac{\alpha_3}{2} < \frac{c}{2^3}, \dots \text{ (by IH)}$$

$$\text{Now, } (\alpha_1 + \alpha_2 + \dots + \alpha_k + \alpha_{k+1}) < c + \frac{c}{2} + \frac{c}{2^2} + \dots + \frac{c}{2^k}$$

$$= \frac{c \left(1 - \frac{1}{2^{k+1}} \right)}{1 - 1/2} = 2c \left(1 - \frac{1}{2^{k+1}} \right) < 2c$$

$$\therefore \alpha_1 + \alpha_2 + \dots + \alpha_{k+1} < 2c$$

$$\Rightarrow -2a(\alpha_1 + \alpha_2 + \dots + \alpha_{k+1}) > -4ac$$

$$\Rightarrow b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_{k+1}) > b^2 - 4ac > 0$$

$\therefore \alpha_{k+2}$ is well defined. Again by IH we have

$$\alpha_{k+1} < \frac{\alpha_k}{2} \Rightarrow 2\alpha_{k+1} < \alpha_k$$

$$\Rightarrow 4\alpha_{k+1}^2 < \alpha_k^2 \text{ [As by def. } \alpha_{k+1}, \alpha_k \text{ are +ve]}$$

$$\Rightarrow 4\alpha_{k+1} < \frac{\alpha_k^2}{\alpha_{k+1}}$$

$$\Rightarrow 4\alpha_{k+1} < \frac{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_k)}{a}$$

$$\Rightarrow 4a\alpha_{k+1} < b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_k)$$

$$\Rightarrow 2a\alpha_{k+1} < b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_k + \alpha_{k+1})$$

$$\Rightarrow \frac{a\alpha_{k+1}^2}{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_{k+1})} < \frac{1}{2}$$

$$\Rightarrow \frac{a\alpha_{k+1}}{b^2 - 2a(\alpha_1 + \alpha_2 + \dots + \alpha_{k+1})} < \frac{\alpha_{k+1}}{2}$$

$$\Rightarrow \alpha_{k+2} < \frac{\alpha_{k+1}}{2}$$

$\therefore P(k+1)$ is also true.

Thus by the Principle of Mathematical Induction the Statement $P(n)$ is true $\forall n \in N$.

27. Let $P(n) : (25)^{n+1} - 24n + 5735$

For $n = 1$.

$P(1) : 625 - 24 + 5735 = 6336 = (24)^2 \times (11)$, which is divisible by 24^2 . Hence $P(1)$ is true

Let $P(k)$ be true, where $k \geq 1$

$$\Rightarrow (25)^{k+1} - 24k + 5735 = (24)^2 \lambda \text{ where } \lambda \in N$$

$$\text{For } n = k+1, P(k+1) : (25)^{k+2} - 24(k+1) + 5735 = 25[(25)^{k+1} - 24k + 5735]$$

$$+ 25 \cdot 24 \cdot k - (25)(5735) + 5735 - 24(k+1)$$

$$= 25(24)^2 \lambda + (24)^2 k - 5735 \times 24 - 24$$

$$= 25(24)^2 \lambda + (24)^2 k - (24)(5736)$$

$$= 25(24)^2 \lambda + (24)^2 k - (24)^2 (239),$$

$$= (24)^2 [25\lambda + k - 239] \text{ which is divisible by } (24)^2.$$

Hence, by the method of mathematical induction result is true $\forall n \in N$.

28. To prove that

$$2^k {}^nC_0 {}^nC_k - 2^{k-1} {}^nC_1 {}^{n-1}C_{k-1} + 2^{k-2} {}^nC_2 {}^{n-2}C_{k-2} - \dots + (-1)^k {}^nC_k {}^{n-k}C_0 = {}^nC_k$$

Mathematical Induction and Binomial Theorem

LHS of above equation can be written as

$$\begin{aligned}
 & \sum_{r=0}^k (-1)^r 2^{k-r} {}^nC_r {}^{n-r}C_{k-r} \\
 &= \sum_{r=0}^k (-1)^r 2^{k-r} \frac{n!}{r!(n-r)!} \frac{(n-r)!}{(k-r)!(n-k)!} \\
 &= \sum_{r=0}^k (-1)^r 2^{k-r} \frac{n!k!}{r!k!(n-k)!(k-r)!} \\
 &= \sum_{r=0}^k (-1)^r \frac{2^k}{2^r} \cdot \frac{n!}{k!(n-k)!} \frac{k!}{r!(k-r)!} \\
 &= 2^k {}^nC_k \sum_{r=0}^k (-1/2)^r \frac{k!}{r!(k-r)!} \\
 &= 2^k {}^nC_k \sum_{r=0}^k {}^kC_r (-1/2)^r = 2^k {}^nC_k (1-1/2)^k \\
 &= 2^k {}^nC_k \frac{1}{2^k} = {}^nC_k = \text{R.H.S. Hence Proved}
 \end{aligned}$$

29. We have $\alpha + \beta = 1 - p$ and $\alpha\beta = -p(1-p)$
For $n = 1, p_n = p_1 = 1$

$$\begin{aligned}
 \text{Also, } A\alpha^n + B\beta^n &= A\alpha + B\beta = \frac{(p^2 + \beta - 1)\alpha}{\alpha\beta - \alpha^2} \\
 &+ \frac{(p^2 + \alpha - 1)\beta}{\alpha\beta - \beta^2} = \frac{p^2 + \beta - 1}{\beta - \alpha} + \frac{p^2 + \alpha - 1}{\alpha - \beta} \\
 &= \frac{p^2 + \beta - 1 - p^2 - \alpha + 1}{\beta - \alpha} = \frac{\beta - \alpha}{\beta - \alpha} = 1
 \end{aligned}$$

For $n = 2, p_2 = 1 - p^2$

$$\begin{aligned}
 \text{Also, } A\alpha^n + B\beta^n &= A\alpha^2 + B\beta^2 \\
 &= \frac{(p^2 + \beta - 1)\alpha^2}{\alpha\beta - \alpha^2} + \frac{(p^2 + \alpha - 1)\beta^2}{\alpha\beta - \alpha^2}
 \end{aligned}$$

which is true for $n = 2$

Now let result is true for $k < n$ where $n \geq 3$.

$$P_n = (1-p)P_{n-1} + p(1-p)P_{n-2}$$

$$\begin{aligned}
 &= (1-p)(A\alpha^{n-1} + B\beta^{n-1}) + p(1-p)(A\alpha^{n-2} + B\beta^{n-2}) \\
 &= A\alpha^{n-2}\{(1-p)\alpha + p(1-p)\} + B\beta^{n-2}\{(1-p)\beta - p(1-p)\}
 \end{aligned}$$

$$\begin{aligned}
 &= A\alpha^{n-2}\{(\alpha + \beta)\alpha - \alpha\beta\} \\
 &\quad + B\beta^{n-2}\{(\alpha + \beta)\beta - \alpha\beta\} \text{ by (1)}
 \end{aligned}$$

$$= A\alpha^{n-2}\{\alpha^2 + \beta\alpha - \alpha\beta\} + B\beta^{n-2}\{\alpha\beta + \beta^2 - \alpha\beta\}$$

$$= A\alpha^{n-2}(\alpha^2) + B\beta^{n-2}(\beta^2) = A\alpha^n + B\beta^n$$

This is true for n . Hence by principle of mathematical induction, the result holds good for all $n \in N$.

I. Integer Value Correct Type

1. (6) Let the coefficients of three consecutive terms of $(1+x)^{n+5}$ be

$${}^{n+5}C_{r-1}, {}^{n+5}C_r, {}^{n+5}C_{r+1}, \text{ then we have } {}^{n+5}C_{r-1} : {}^{n+5}C_r : {}^{n+5}C_{r+1} = 5 : 10 : 14$$

$$\frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_r} = \frac{5}{10} \Rightarrow \frac{r}{n+6-r} = \frac{1}{2}$$

$$\text{or } n - 3r + 6 = 0 \quad \dots(1)$$

$$\text{Also } \frac{{}^{n+1}C_r}{{}^{n+5}C_{r+1}} = \frac{10}{14} \Rightarrow \frac{r+1}{n-r+5} = \frac{5}{7}$$

$$\text{or } 5n - 12r + 18 = 0 \quad (2)$$

Solving (1) and (2) we get $n = 6$.

2. (5) $(1+x)^2 + (1+x)^3 + \dots + (1+x)^{49} + (1+mx)^{50}$

$$= (1+x)^2 \left[\frac{(1+x)^{48} - 1}{(1+x) - 1} \right] + (1+mx)^{50}$$

$$= \frac{1}{x} \left[(1+x)^{50} - (1+x)^2 \right] + (1+mx)^{50}$$

Coeff. of x^2 in the above expansion

$$= \text{Coeff. of } x^3 \text{ in } (1+x)^{50} + \text{Coeff. of } x^2 \text{ in } (1+mx)^{50}$$

$$\Rightarrow {}^{50}C_3 + {}^{50}C_2 m^2$$

$$\therefore (3n+1) {}^{51}C_3 = {}^{50}C_3 + {}^{50}C_2 m^2$$

$$\Rightarrow (3n+1) = \frac{{}^{50}C_3}{{}^{51}C_3} + \frac{{}^{50}C_2}{{}^{51}C_3} m^2$$

$$\Rightarrow 3n+1 = \frac{16}{17} + \frac{1}{17} m^2 \Rightarrow n = \frac{m^2 - 1}{51}$$

Least positive integer m for which n is an integer is $m = 16$ and then $n = 5$

Section-B

JEE Main/ AIEEE

1. (a) We have $t_{p+1} = {}^{p+q}C_p x^p$ and $t_{q+1} = {}^{p+q}C_q x^q$
 ${}^{p+q}C_p = {}^{p+q}C_q$. [Remember ${}^nC_r = {}^nC_{n-r}$]
 2. (c) We have $2^n = 4096 = 2^{12} \Rightarrow n = 12$; the greatest coeff
 = coeff of middle term. So middle term

$$= t_7; t_7 = t_{6+1} \Rightarrow \text{coeff of } t_7 = {}^{12}C_6 = \frac{12!}{6!6!} = 924.$$

3. (d) $(1 + 0.0001)^{10000} = \left(1 + \frac{1}{n}\right)^n, n = 10000$

$$= 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{2!} \cdot \frac{1}{n^2} + \frac{n(n-1)(n-2)}{3!} \cdot \frac{1}{n^3} + \dots$$

$$= 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) + \left(1 - \frac{2}{n}\right) + \dots$$

$$< 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{(9999)!}$$

$$= 1 + \frac{1}{1!} + \frac{1}{2!} + \dots \infty = e < 3$$

4. (c) $t_{r+2} = {}^{2n}C_{r+1} x^{r+1}; t_{3r} = {}^{2n}C_{3r-1} x^{3r-1}$
 Given ${}^{2n}C_{r+1} = {}^{2n}C_{3r-1}$;
 $\Rightarrow {}^{2n}C_{2n-(r+1)} = {}^{2n}C_{3r-1}$
 $\Rightarrow 2n - r - 1 = 3r - 1 \Rightarrow 2n = 4r \Rightarrow n = 2r$

5. (b) $a_1 = \sqrt{7} < 7$. Let $a_m < 7$

Then $a_{m+1} = \sqrt{7 + a_m} \Rightarrow a_{m+1}^2 = 7 + a_m < 7 + 7 < 14$.
 $\Rightarrow a_{m+1} < \sqrt{14} < 7$; So by the principle of
 mathematical induction $a_n < 7 \forall n$.

6. (d) $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} (x)^r$

For first negative term, $n - r + 1 < 0 \Rightarrow r > n + 1$

$$\Rightarrow r > \frac{32}{5} \therefore r = 7. \left(\because n = \frac{27}{5} \right)$$

Therefore, first negative term is T_8 .

7. (c) $T_{r+1} = {}^{256}C_r (\sqrt{3})^{256-r} (\sqrt[8]{5})^r = {}^{256}C_r (3)^{\frac{256-r}{2}} (5)^{r/8}$

Terms will be integral if $\frac{256-r}{2}$ & $\frac{r}{8}$ both are +ve

integer, which is so if r is an integral multiple of 8. As
 $0 \leq r \leq 256$

$\therefore r = 0, 8, 16, 24, \dots, 256$, total 33 values.

8. (b) $S(k) = 1 + 3 + 5 + \dots + (2k-1) = 3 + k^2$

$S(1) : 1 = 3 + 1$, which is not true

$\therefore S(1)$ is not true.

$\therefore$ P.M.I cannot be applied

Let $S(k)$ is true, i.e.

$$1 + 3 + 5 + \dots + (2k-1) = 3 + k^2$$

$$\Rightarrow 1 + 3 + 5 + \dots + (2k-1) + 2k + 1$$

$$= 3 + k^2 + 2k + 1 = 3 + (k+1)^2$$

$$\therefore S(k) \Rightarrow S(k+1)$$

9. (c) The middle term in the expansion of

$$(1 + \alpha x)^4 = T_3 = {}^4C_2 (\alpha x)^2 = 6\alpha^2 x^2$$

The middle term in the expansion of

$$(1 - \alpha x)^6 = T_4 = {}^6C_3 (-\alpha x)^3 = -20\alpha^3 x^3$$

According to the question

$$6\alpha^2 = -20\alpha^3 \Rightarrow \alpha = -\frac{3}{10}$$

10. (b) Coeff of x^n in $(1+x)(1-x)^n$

$$= \text{Coeff of } x^n \text{ in } (1-x)^n + \text{Coeff of } x^{n-1} \text{ in } (1-x)^n$$

$$= (-1)^n {}^nC_n + (-1)^{n-1} {}^nC_{n-1} = (-1)^n 1 + (-1)^{n-1} n$$

$$= (-1)^n [1 - n]$$

11. (d) ${}^{50}C_4 + \sum_{r=1}^6 {}^{56-r}C_3$

$$\Rightarrow {}^{50}C_4 + \left[{}^{55}C_3 + {}^{54}C_3 + {}^{53}C_3 + {}^{52}C_3 + {}^{51}C_3 + {}^{50}C_3 \right]$$

$$\text{We know } \left[{}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r \right]$$

$$\Rightarrow ({}^{50}C_4 + {}^{50}C_3)$$

$$+ {}^{51}C_3 + {}^{52}C_3 + {}^{53}C_3 + {}^{54}C_3 + {}^{55}C_3$$

$$\Rightarrow ({}^{51}C_4 + {}^{51}C_3) + {}^{52}C_3 + {}^{53}C_3 + {}^{54}C_3 + {}^{55}C_3$$

Proceeding in the same way, we get

$$\Rightarrow {}^{55}C_4 + {}^{55}C_3 = {}^{56}C_4.$$

12. (a) We observe that

$$A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \text{ and we can prove by}$$

$$\text{induction that } A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Now } nA - (n-1)I &= \begin{bmatrix} n & 0 \\ n & n \end{bmatrix} - \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} = A^n \end{aligned}$$

$$\therefore nA - (n-1)I = A^n$$

13. (d) T_{r+1} in the expansion

$$\left[ax^2 + \frac{1}{bx} \right]^{11} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx} \right)^r$$

$$= {}^{11}C_r (a)^{11-r} (b)^{-r} (x)^{22-2r-r}$$

For the Coefficient of x^7 , we have

$$\Rightarrow 22 - 3r = 7 \Rightarrow r = 5$$

$$\therefore \text{Coefficient of } x^7 = {}^{11}C_5 (a)^6 (b)^{-5} \dots (1)$$

Again T_{r+1} in the expansion

$$\left[ax - \frac{1}{bx^2} \right]^{11} = {}^{11}C_r (ax)^{11-r} \left(-\frac{1}{bx^2} \right)^r$$

$$= {}^{11}C_r (a)^{11-r} (-1)^r \times (b)^{-r} (x)^{-2r} (x)^{11-r}$$

For the Coefficient of x^{-7} , we have

$$\text{Now } 11 - 3r = -7 \Rightarrow 3r = 18 \Rightarrow r = 6$$

$$\therefore \text{Coefficient of } x^{-7} = {}^{11}C_6 a^5 \times 1 \times (b)^{-6}$$

$$\therefore \text{Coefficient of } x^7 = \text{Coefficient of } x^{-7}$$

$$\Rightarrow {}^{11}C_5 (a)^6 (b)^{-5} = {}^{11}C_6 a^5 \times (b)^{-6} \Rightarrow ab = 1.$$

14. (c) $\therefore x^3$ and higher powers of x may be neglected

$$\therefore \frac{(1+x)^{\frac{3}{2}} - \left(1 + \frac{x}{2}\right)^3}{\left(1 - x^2\right)}$$

$$= (1-x)^{\frac{-1}{2}} \left[\left(1 + \frac{3}{2}x + \frac{\frac{3}{2} \cdot \frac{1}{2}}{2!} x^2\right) - \left(1 + \frac{3x}{2} + \frac{3 \cdot 2}{2!} \frac{x^2}{4}\right) \right]$$

$$= \left[1 + \frac{x}{2} + \frac{\frac{1}{2} \cdot \frac{3}{2}}{2!} x^2 \right] \left[\frac{-3}{8} x^2 \right] = \frac{-3}{8} x^2$$

(as x^3 and higher powers of x can be neglected)

15. (d) $(1-ax)^{-1} (1-bx)^{-1}$

$$= (1+ax+a^2x^2+\dots)(1+bx+b^2x^2+\dots)$$

$\therefore$ Coefficient of x^n

$$x^n = b^n + ab^{n-1} + a^2b^{n-2} + \dots + a^{n-1}b + a^n$$

{which is a G.P. with $r = \frac{a}{b}$ }

$$\therefore \text{Its sum is } = \frac{b^n \left[1 - \left(\frac{a}{b} \right)^{n+1} \right]}{1 - \frac{a}{b}} = \frac{b^{n+1} - a^{n+1}}{b-a}$$

$$\therefore a_n = \frac{b^{n+1} - a^{n+1}}{b-a}$$

16. (d) $(1-y)^m (1+y)^n$

$$= [1 - {}^mC_1 y + {}^mC_2 y^2 - \dots]$$

$$[1 + {}^nC_1 y + {}^nC_2 y^2 + \dots]$$

$$= 1 + (n-m) + \left\{ \frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn \right\} y^2 + \dots$$

$$\therefore a_1 = n - m = 10$$

$$\text{and } a_2 = \frac{m^2 + n^2 - m - n - 2mn}{2} = 10$$

$$\text{So, } n - m = 10 \text{ and } (m-n)^2 - (m+n) = 20$$

$$\Rightarrow m + n = 80$$

$$\therefore m = 35, n = 45$$

17. (b) $T_{r+1} = (-1)^r \cdot {}^nC_r (a)^{n-r} \cdot (b)^r$ is an expansion of $(a-b)^n$

$$\therefore \text{5th term} = t_5 = t_{4+1} = (-1)^4 \cdot {}^nC_4 (a)^{n-4} \cdot (b)^4 = {}^nC_4 \cdot a^{n-4} \cdot b^4$$

$$\text{6th term} = t_6 = t_{5+1} = (-1)^5 {}^nC_5 (a)^{n-5} (b)^5$$

$$\text{Given } t_5 + t_6 = 0$$

$$\therefore {}^nC_4 \cdot a^{n-4} \cdot b^4 + (-1)^5 {}^nC_5 \cdot a^{n-5} \cdot b^5 = 0$$

$$\Rightarrow \frac{n!}{4!(n-4)!} \cdot \frac{a^n}{a^4} \cdot b^4 - \frac{n!}{5!(n-5)!} \cdot \frac{a^n b^5}{a^5} = 0$$

$$\Rightarrow \frac{n! \cdot a^n b^4}{4!(n-4)! \cdot a^4} \left[\frac{1}{(n-4)} - \frac{b}{5a} \right] = 0$$

$$\text{or, } \frac{1}{n-4} - \frac{b}{5a} = 0 \Rightarrow \frac{a}{b} = \frac{n-4}{5}$$

18. (d) We know that, $(1+x)^{20} = {}^{20}C_0 + {}^{20}C_1 x + {}^{20}C_2 x^2 + \dots$

$${}^{20}C_{10} x^{10} + \dots + {}^{20}C_{20} x^{20}$$

$$\text{Put } x = -1, (0) = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots + {}^{20}C_{10}$$

$$- {}^{20}C_{11} + \dots + {}^{20}C_{20}$$

$$\Rightarrow 0 = 2[{}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - {}^{20}C_9] + {}^{20}C_{10}$$

$$\Rightarrow {}^{20}C_{10} = 2[{}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots - {}^{20}C_9 + {}^{20}C_{10}]$$

$$\Rightarrow {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \dots + {}^{20}C_{10} = \frac{1}{2} {}^{20}C_{10}$$

19. (b) We have

$$\sum_{r=0}^n (r+1) {}^nC_r x^r = \sum_{r=0}^n r \cdot {}^nC_r x^r + \sum_{r=0}^n {}^nC_r x^r$$

$$= \sum_{r=1}^n r \cdot \frac{n}{r} \cdot {}^{n-1}C_{r-1} x^r + (1+x)^n$$

$$= nx \sum_{r=1}^n {}^{n-1}C_{r-1} x^{r-1} + (1+x)^n$$

$$= nx(1+x)^{n-1} + (1+x)^n = \text{RHS}$$

∴ Statement 2 is correct.

Putting $x = 1$, we get

$$\sum_{r=0}^n (r+1)^n C_r = n \cdot 2^{n-1} + 2^n = (n+2) \cdot 2^{n-1}.$$

∴ Statement 1 is also true and statement 2 is a correct explanation for statement 1.

20. (a) $(8)^{2n} - (62)^{2n+1}$
 $= (64)^n - (62)^{2n+1} = (63+1)^n - (63-1)^{2n+1}$
 $= \left[{}^nC_0 (63)^n + {}^nC_1 (63)^{n-1} + {}^nC_2 (63)^{n-2} \right.$
 $\quad \left. + \dots + {}^nC_{n-1} (63) + {}^nC_n \right]$
 $= \left[{}^{2n+1}C_0 (63)^{2n+1} - {}^{2n+1}C_1 (63)^{2n} + {}^{2n+1}C_2 (63)^{2n-1} \right.$
 $\quad \left. - \dots + (-1)^{2n+1} {}^{2n+1}C_{2n+1} \right]$
 $= 63 \times \left[{}^nC_0 (63)^{n-1} + {}^nC_1 (63)^{n-2} + {}^nC_2 (63)^{n-3} \right.$
 $\quad \left. + \dots \right] + 1$
 $- 63 \times \left[{}^{2n+1}C_0 (63)^{2n} - {}^{2n+1}C_1 (63)^{2n-1} + \dots \right] + 1$
 $\Rightarrow 63 \times \text{some integral value} + 2$
 $\Rightarrow 8^{2n} - (62)^{2n+1}$ when divided by 9 leaves 2 as the remainder.

21. (b) $S_2 = \sum_{j=1}^{10} j {}^{10}C_j = \sum_{j=1}^{10} 10 {}^9C_{j-1}$

$$= 10 \left[{}^9C_0 + {}^9C_1 + {}^9C_2 + \dots + {}^9C_9 \right] = 10 \cdot 2^9$$

22. (b) $(1-x-x^2+x^3)^6 = [(1-x)-x^2(1-x)]^6$
 $= (1-x)^6 (1-x^2)^6$
 $= (1-6x+15x^2-20x^3+15x^4-6x^5+x^6)$
 $\times (1-6x^2+15x^4-20x^6+15x^8-6x^{10}+x^{12})$
Coefficient of $x^7 = (-6)(-20) + (-20)(15) + (-6)(-6)$
 $= -144$

23. (a) $(\sqrt{3}+1)^{2n} - (\sqrt{3}-1)^{2n}$
 $= \left[(\sqrt{3}+1)^2 \right]^n - \left[(\sqrt{3}-1)^2 \right]^n$
 $= (4+2\sqrt{3})^n - (4-2\sqrt{3})^n$
 $= 2^n \left[(2+\sqrt{3})^n - (2-\sqrt{3})^n \right]$

$$= 2^n \times 2 \left[{}^nC_1 2^{n-1} \sqrt{3} + {}^nC_3 \cdot 2^{n-3} 3\sqrt{3} + \dots \right]$$

$$= 2^{n+1} \sqrt{3} \left[{}^nC_1 \cdot 2^{n-1} + {}^nC_3 2^{n-3} \cdot 3 + \dots \right]$$

$$= \sqrt{3} \times \text{Some integer} \therefore \text{irrational number}$$

24. (c) Given expression can be written as

$$\left((x^{1/3} + 1) - \left(\frac{\sqrt{x} + 1}{\sqrt{x}} \right) \right)^{10} = \left(x^{1/3} + 1 - 1 - \frac{1}{\sqrt{x}} \right)^{10}$$

$$= (x^{1/3} - x^{-1/2})^{10}$$

$$\text{General term} = T_{r+1} = {}^{10}C_r (x^{1/3})^{10-r} (-x^{-1/2})^r$$

$$= {}^{10}C_r x^{\frac{10-r}{3}} \cdot (-1)^r \cdot x^{-\frac{r}{2}} = {}^{10}C_r (-1)^r \cdot x^{\frac{10-r}{3} - \frac{r}{2}}$$

$$\text{Term will be independent of } x \text{ when } \frac{10-r}{3} - \frac{r}{2} = 0$$

$$\Rightarrow r = 4$$

$$\text{So, required term} = T_5 = {}^{10}C_4 = 210$$

25. (b) Consider $(1+ax+bx^2)(1-2x)^{18}$
 $= (1+ax+bx^2) [{}^{18}C_0 - {}^{18}C_1(2x) + {}^{18}C_2(2x)^2 - {}^{18}C_3(2x)^3$
 $\quad + {}^{18}C_4(2x)^4 - \dots]$
Coeff of $x^3 = {}^{18}C_3(-2)^3 + a \cdot (-2)^2 \cdot {}^{18}C_2 + b(-2) \cdot {}^{18}C_1 = 0$
Coeff. of $x^3 = -{}^{18}C_3 \cdot 8 + a \times 4 \cdot {}^{18}C_2 - 2b \times 18 = 0$
 $= -\frac{18 \times 17 \times 16}{6} \cdot 8 + \frac{4a + 18 \times 17}{2} - 36b = 0$
 $= -51 \times 16 \times 8 + a \times 36 \times 17 - 36b = 0$
 $= -34 \times 16 + 51a - 36b = 0$
 $= 51a - 36b = 34 \times 16 = 544$
 $= 51a - 36b = 544 \quad \dots(i)$
Only option number (b) satisfies the equation number (i)

26. (c) $(1-2\sqrt{x})^{50} = {}^{50}C_0 - {}^{50}C_1 2\sqrt{x} + {}^{50}C_2 (2\sqrt{x})^2 \dots (1)$

$$(1+2\sqrt{x})^{50} = {}^{50}C_0 + {}^{50}C_1 2\sqrt{x} - {}^{50}C_2 (2\sqrt{x})^2$$

$$+ \dots + {}^{50}C_3 (2\sqrt{x})^3 - {}^{50}C_4 (2\sqrt{x})^4 \dots (2)$$

Adding equation (1) and (2)

$$(1-2\sqrt{x})^{50} + (1+2\sqrt{x})^{50}$$

$$= 2 \left[{}^{50}C_0 + {}^{50}C_2 2^2 x + {}^{50}C_4 2^4 x^2 + \dots \right]$$

$$\text{Putting } x = 1, \text{ we get above as } \frac{3^{50} + 1}{2}$$

27. (b) Total number of terms $= n+2 = {}^nC_2 = 28$
 $(n+2)(n+1) = 56$
 $x = 6$
Sum of coefficients $= (1-2+4)^n = 3^6 = 729$